Master Thesis

Development of Driving Vision-Workload Measures to Evaluate Safety Level

by

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Abstract

As previous traffic research regarding workload and safety did not look at visual physiology and performance, the thesis investigated recent attempts to integrate these measures as dependent variables. The thesis then examined visual measures to develop them in light of complex applications.

It has been suggested that the pupil dilates and blinks eyes during cognitive effort and the degree of dilation and blinking depends on the difficulty level. Additionally, various video-based eye trackers using cameras and infrared illuminators were discussed.

The literature review introduces how research in vision studies cognitive load through gaze behavior, pupillometry, and blinks. An investigation of each research was conducted based on a range of criteria for study inclusion after completing a comprehensive literature search. To meet the inclusion criteria, it was required to quantify the independent variable of cognitive workload and the dependent variable of vision behavior measurement. Using restricted criteria, driving vision-workload measurements candidate list was compiled.

The study selected the number of fixation visits and fixation duration to differentiate vision workload between complex sign design and simple sign design. Next, the thesis compared tangent and curve segment pupillometry measurements without signs to determine whether pupillometry measures and fixation indicators

were valid within geographical workloads assessment.

We also compared a regular roadway exposed to surfaces with an underpass to determine the validity of pupillometry measures regarding mental workload regarding environmental surroundings.

Research using a driving simulator has been considered a valid tool for evaluating countermeasures related to driving behavior since it minimizes confounding effects by controlling the various factors affecting driving behavior. A driving simulator study is typically safer and less expensive than a field study, and data collection is relatively easy. The results of the driving simulation study were analyzed objectively. Researchers recruited 37 subjects for the experiment; one could not participate due to nausea.

This research contributes to the body of knowledge by showing That a modular framework is presented for analyzing, defining, measuring, and applying vision workload in various contexts based on multidimensional phenomenon theory.

Keyword: Driving simulator, Eye tracker, Driving workload, Visual Behavior, Driving Behavior

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Chapter 1 Introduction

1. Background

In recent decades, there has been an increase in studies exploring the relationship between vision and driving. This topic has gained increased attention due to the need for evidence-based standards for determining vision standards for driving licenses and a lack of knowledge about how vision impairment affects driver performance and safety. In addition to on-road and knowledge tests, visual acuity testing is the most common way to determine eligibility for licensure worldwide. Despite this, little to no evidence shows that visual acuity screening tests improve driver performance or safety, regardless of the selected pass-fail cut-point. The lack of evidence-based vision standards for licensure has prompted increasing interest in vision and driving among clinicians and researchers. Therefore, clinicians and policymakers must understand how different study designs and measurement methods can be interpreted so that they can make accurate conclusions and recommendations. Various assessment tools are reviewed in this study, and implications of previous studies are discussed regarding vision and driving (e.g., driver safety, performance, and subjective reports).

2. Objective

This study was designed to bridge the gap between the qualitative methods currently used for assessing ocular behavior in vision research and the quantitative that are desired for evaluating mental workload while driving. It will demonstrate how various vision properties, specifically physiological indicators, can be used to derive mental workload in driving research. The power of the vision workload concept is that one can measure an exact quantity of "workload" present. In this research, the ultimate goal was to develop and validate the vision workload methodology using eye movements by specifying a range of data before correlation with performance.

The main contribution of this thesis is the introduction of a methodology developed as a formal framework to represent vision workload as a dynamic and multidimensional phenomenon and to assess it as a usable numerical index. This research contributes to the body of knowledge by providing a modular framework built upon multidimensional phenomenon theory in which vision workload can be robustly modeled, defined, measured, and applied in different contexts.

3. Study Scope and Flow

For this thesis, The experiment investigated the validity of each selected physiological indicator of vision workload in each case study to show potential development as a formal framework to represent vision demand in driving research. The methods described in Research synthesis and Meta-analysis: A step-by-step Approach by Harris Cooper (2010) was applied to the thesis. Cooper summarizes the meta-analysis process into seven steps: 1) formulating the problem, 2) searching the literature, 3) gathering information from studies, 4) evaluating the quality of studies, 5) analyzing and integrating the outcomes of studies, 6) interpreting the evidence, and 7) presenting the results as Figure 1 shown. While following those steps, this thesis is divided into five further Chapters:

Chapter 2 The literature review introduces how research in vision studies cognitive load through gaze behavior, pupillometry, and blinks. A description of how vision researchers have used velocity and dispersion to categorize fixation, and saccadic movement was provided.

Chapter 3 describe the methodology and design for this research, including how the aim and objectives are addressed and the driving simulator experiment and reasoning;

Chapter 4 describes the process of application and validation, including a comparison of the database used for further analysis; Through explaining the process, a development of measures was

provided.

Chapter 5 summarizes the main findings of this study, including the way eye movement measurements can provide insight into the relationship between vision and cognitive workload, as well as future areas for study.

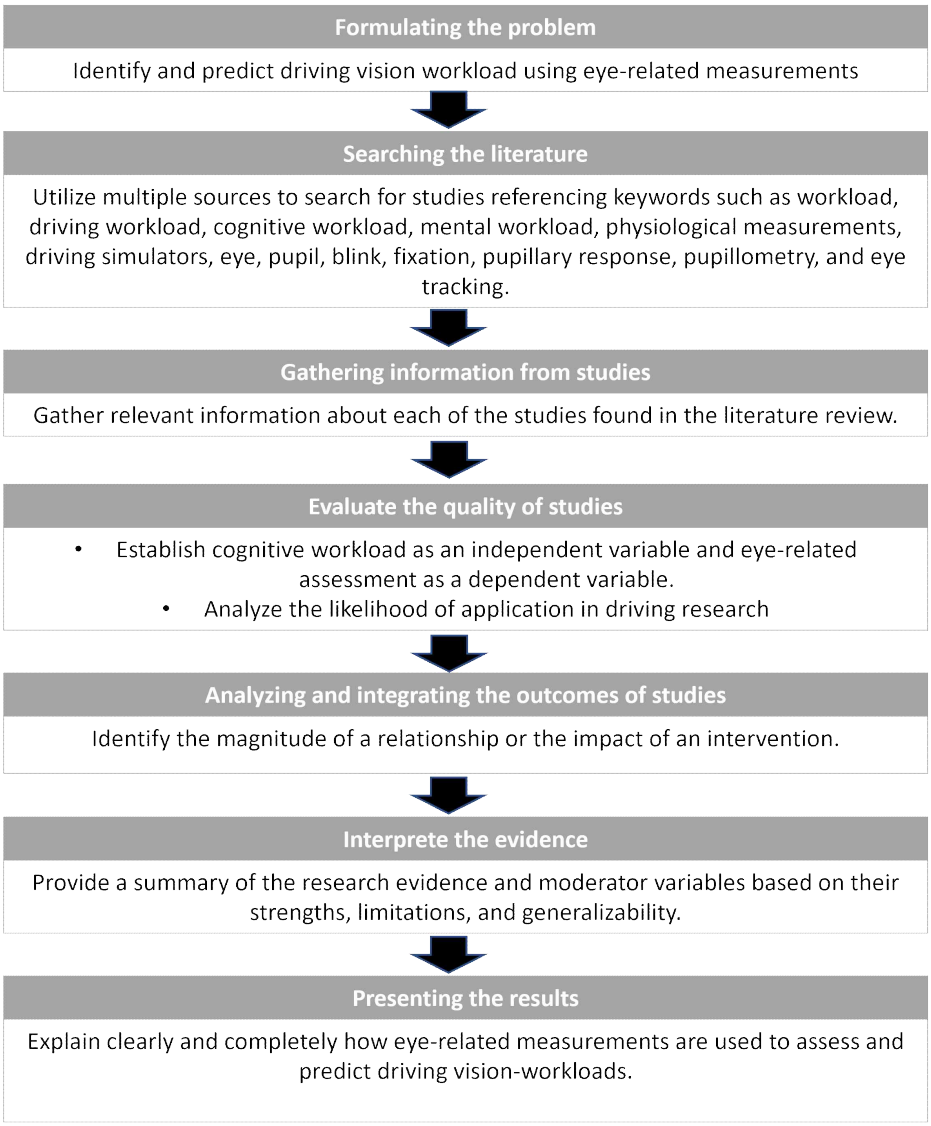


Figure 1. Seven steps flowchart of thesis

Chapter 2 Literature Review

There are six sections to this literature review, the first of which presents a definition of workload, a review of existing models to describe the relationship between workload and performance, and between behavior and performance. Driver coping strategies and their impact on workload and driving performance have been reviewed. Next, the second section describes existing studies on driving simulators, how they relate to workload and behavior research, and how they might be used to understand cognitive demand and performance better. The driving simulator research review included research environment consistency and safety measures and their application to driving and visual behavior. During the third section, key findings regarding driving workload and gaze performance are presented. The fourth and fifth sections present pupillometry and blink studies, along with potential research methods for measuring driving vision-workload. In the final section of this chapter, we summarize these discussions and present our conclusions.

1. Measures of effectiveness for Vision-Workload

1.1 Mental Workload

The study of stress has been around for many years in psychology, originating from some of the earliest works on learning under compulsion. Chronic stress has long been

recognized as a trigger for illness and debilitation. Like the British industrial fatigue research board, fatigue has been a significant topic in industrial research since the turn of the century, providing early insights into disciplines such as occupational medicine, industrial hygiene, and systems safety that have been seminal today (Hancock, P. A., & Desmond, P. A, 2001). On the other hand, the mental workload is relatively new, emerging as aviation psychologists are confronted with ever more loaded and taxed pilots at the controls of high-performance aircraft. To identify and predict human performance responses, we must go beyond traditional information-processing models' sterile, mathematical formulations to incorporate emotive, practical, or more generally energetic factors.

It is essential to recognize that the changes made to the psycho-physical task parameters in these studies moderated workload but did not significantly reduce stress measurements. Some researchers indicated that participants often report increased stress levels after completing vigilance tasks, suggesting that other factors contribute to the perception of workload (Galinsky, Rosa, Warm, & Dember, 1993), and others have shown that vigilance can induce physiological changes associated with stress (Frankenhaeuser, Nordheden, Myrsten,& Post, 1971). Furthermore, Hancock & Warm (1989) have argued that maintaining high levels of vigilance performance may be sufficient to produce a stress response. As defined by Hancock and Warm (1989), psychological

and physical stability require an optimal degree of both information structure and rate of information. Most vigilance operators, however, need help to control information structure or rate. Thus, an optimal level can only be achieved or maintained with control over these factors. Parasuraman (1985) showed that higher event rates demand more attentional resources, and Galinsky, Dember, and Warm (1989) found that workload scores on the TLX were significantly higher with higher event rates.

The workload must not be viewed as a static phenomenon or as a factor solely determined by the tasks imposed on a driver but rather as a dynamic and multidimensional phenomenon, depending on whether the driver is initiating changes, the task at hand, the driving environment, or a combination of those factors. The basis of this perspective lies in the concept that drivers are goal-oriented and purposeful (Craik, 1947; Newell &Simon, 1972). In this perspective, mental workload represents the degree to which humans and machines collaborate effectively and synergistically.

Despite our best efforts, traffic accidents are numerous, seemingly ineradicable, costly, and primarily caused by humans. Most road accidents are caused by human errors in poor perception, insufficient attention, and inadequate information processing (Smiley& Brookhuis, 1987; Treat et al., 1977). Even though low and high mental workloads may contribute to these errors, an exact relation between mental workload and accident

causation needs to be established or measured. According to Brookhuis, VanWinsum, Heijer, and Duynstee (1999), underload leads to lowered alertness and attention, whereas overload leads to distraction, diverted attention, and inadequate time for adequately processing information. It has been studied that both factors are associated with driver impairment, but there is no direct connection between them and accident causation (Brookhuis et al., 2003). It is necessary to establish criteria for determining when impairment falls below a specific threshold, which leads to accidents. Accidents and mental workload can only be linked when the origins, such as information overload, boredom, fatigue, or alcohol or drug abuse, are also examined. In this type of research, driving simulators are likely to be the most effective tool, and some even believe it should only be conducted with driving simulators (Brookhuis & Waard, 2010). To further develop the thesis, the study will review previous workload assessment tools and introduce various attempts at driving.

1.2 Driving Workload

Several studies have argued that vision is the most necessary resource available to drivers and is the essential input for information processing. The limitations of the human visual system have led to numerous studies trying to measure the impact of workload on an individual's visual performance while driving. Homemaker (2002) defines workload as the number of resources

required to perform several tasks concurrently, along with the number of resources required to accomplish them. The author distinguishes three types of driving workload: visual (e.g., for many types of sources, have the driver look?), motor (e.g., what should the driver do with their hands and feet?), and mental (e.g., how many types of information has the driver to process?). Similarly, Hart and Staveland (1988) defined driving workload as a hypothetical construct representing the costs incurred by drivers to achieve a specific level of performance. According to Boer (2005), the mental workload is the effort required to maintain a driving state within a personal safety zone. Drivers must ensure they have adequate resources to assess their visual environment and maintain their lateral and longitudinal positions (Boyce, 1981). Visual information acquisition (scanning); perceptual-motor coordination; anticipating and assessing traffic conditions; estimating risk; setting safety margins; and balancing speed and caution have been identified (Duncan, Williams, & Brown, 1991). Drivers with experience automate many of these processes to perform their multiple tasks under normal conditions (Handcock & Parasuranman, 1992). Approximately 90% of the information received by drivers is visual (Sabey & Staughton, 1975). Vision has been identified as the most important source of information for drivers (Wierwille, 1993), and overstimulation of the visual pathway has been identified as an essential accident prevention research topic (Wierwille, 1993). Wierwille (1993) also explains that forward visual scenes are the

primary information-gathering sources. Drivers' perceptual resources are being overloaded by many of today's traffic conditions, according to Connoly (1968). The introduction of advanced driver information systems will further impair drivers' ability to handle road infrastructure and environmental conditions.

Measuring task performance, subjective reports, and physiological data are the three categories distinguished in traffic research. The first and most used category of measures involves direct registration of the operator's capability to perform a driving task at an acceptable level with a low accident probability. These task performance measures are directly associated with vehicle handling, such as steering and following, as well as lateral and longitudinal vehicle control.

Observer reports, which experts typically provide, and driver reports, which the driver themselves provides, are two types of subjective driving performance reports. For researchers, subjective measurement techniques are easy to use and cost-effective (Endsley, 1995). As well as being multidimensional, they can be used to predict the workload related to proposed systems and designs (Wierwille & Eggemeier, 1993). However, De Rivercourt et al., (2008) concluded that subjective measures could not distinguish task demands from invested efforts. A long delay between subjective ratings and task completion can lead to critical information loss (Wierwille & Eggemeier, 1993). Considering the core of this article is based on physiological indicators related to

vision, the performance measure and subjective reports related to driving tasks will be briefly discussed to conform to the previous approach for assessing mental workload. As shown in Table 1, Paxion (2014) investigated how each measurement tool aligns with the other to compare the advantages of subjective report methods.

Table 1. Advantages of workload questionnaires(Paxion, J et at., 2014)

Advantages	SWAT	WP	NASA-TLX	RSME	DALI
Complementary dimensions of workload	●	●	●		●
Multidimentional workload: Task demand, effort and performance			●		●
Assess workload in a multi-modal system (visual, auditory, etc.)		●			●
Sensitive to the task difficulty		●		●	
Assesses the subjective cost to perform a task			●	●	
Workload comparison for tasks of different difficulty		●			
Predicts the task performance			●		
Analyzes the cognitive demands for a task	●	●			
Used in real complex tasks	●	●	●		
Sensitivity	●	●	●		
Diagnosticity	●	●	●		
Selectivity/Validity	●	●	●		
Intrusiveness	●	●	●		

SWAT:Subjective Workload Assessment Technique
WP:Workload Prifile
NASA-TLX: the NASA Task Load Index
RSME: Evaluation of Rating Scale Mental Effort
DALI: The driving activity load index

Physiological measures are the most natural form of mental workload indices since mental work also requires physiological activity. In addition to the physical workload, the mental workload has a noticeable impact on heart rate variability (Mulder, 1988). As a result of mental workload, heart rate variability might decrease while heart rate increases. The activity of the brain (Brookhuis, 1993; De Warrd & Brookhuis, 1991) and specific muscles of the face(Mesken, 2002) are other measures of interest. Physiological measures can advantageously complement subjective data. As a result, they provide a continuous on-line assessment that responds relatively quickly to phasic shifts in mental workload, even though reaction latency is dependent on the measures used. As such, they provide indirect measures of mental workload. Moreover, they are non-intrusive and can be applied to operational environments with controlled factors that could affect the signal, such as temperature and lighting. Furthermore, they can be used as a baseline in the absence of behavioral activities, providing a fine-grained assessment of the dimensions of mental workload. Although physiological measurements have their advantages, they also come with their disadvantages. For these measurements, recording techniques are significantly more expensive than performance and subjective measures, and it is much harder to distinguish between signal and noise (Wendt, 2018). However, physiological measures can far outweigh their disadvantages. In addition, it can record measurements even when behavior is absent and offers measures

that can respond quickly to changes in workload (Kramer, 1990). Furthermore, these measures are relatively unobtrusive and multidimensional (Kramer, 1990). (Endsley & Garland, 2000) Physiological measurements can provide an objective assessment of cognitive workload. Additionally, systems can continuously assess cognitive workload using physiological measurements, even when task performance does not change (Haapalainen et al., 2010). As a result of the ability to obtain subjective responses or transform performance responses without intervention, this type of measurement is often more attractive as an assessment approach than other types (Marquart, Cabrall, & de Winter, 2015). Table 2 illustrates the advantages of physiological measures over subjective and performance measures. The following physiological measures are most commonly used to measure mental workload objectively.

Table 2. Advantages and Disadvantages of mental workload measurements

Measurement type	Advantages	Disadvantages
Performance	Objective	Indirect insignts about workload
	Nonintrusive	Insensitive sometimes to variations in workload at low tomoderate levels of demand
Subjective	Low cost	Difficulties distinguishing task demands from invested effort
	Ease of use	indirect insights about workload
	Multidimensional	Loss of critical rating information with extended delays
	Predictive assessment Capabilities	
Physiological	Direct measurements over time	Difficulty discriminating between signal and noise
	Recorded in the absense of behavior	
	Unobtrusive	
	Responds quickly to shifts in workload	Expensive recording technique
	Multidimensional	
	Unbiased	

1.2.1 The Electrocardiogram (ECG)

The Electrocardiogram records the heart's electrical activity, which is necessary for cardiac muscle contraction. Mental workload is detected by two main measures (Mulder, 1986, 1988, 1992; de Waard & Brookhuis, 1991; Wilson & Eggemeier, 1991; Jorna, 1992; Boutcher & Boutcher, 2006; Brookhuis and De Waard, 2010; Causse et al., 2011). Firstly, mean heart rate refers to the number of beats per minute, also known as a differential heart rate, used to calculate the difference between rest and activity periods. HR

Variability (HRV) is defined by the variance of Inter Beat Interval, which is the interval between two consecutive peaks. There are several frequencies of HRV, and the mid-frequency band (0.10.0Hz component of HRV) is explicitly used to identify the level of mental effort. Measures such as these, however, have some limitations. Besides the energetic, thermoregulatory, respiratory, and emotional functions (Nickel & Nachreiner, 2003), HR also reflects physical activity and emotional strain(Jahn et al., 2005). According to Gabaude et al., (2012), HRV does not always distinguish difficulty levels.

1.2.2 The Electroencephalogram(EEG)

In the electroencephalogram, two types of indicators are recorded: bands of frequencies and event-related potentials. In terms of the bands, a decrease in the alpha band (8 to 13 Hz) and an increase in theta band (4 to 8 Hz) suggest an increase in mental workload (Kramer, 1991; Borghini et al., 2014), although further research is needed to confirm this link(de Waard, 1996). ERPs provide a picture of mental chronometry, particularly in identifying perceptual, cognitive, and motor processes involved in complex situations. Many long-latency ERP components (positive or negative potentials occurring 100, 200, or 300 ms after stimulus presentation) are considered. Despite their high diagnostic accuracy, ERPs are insensitive to response factors, as other electrical signals influence them and have a poor signal-to-noise ratio.

1.2.3 The Electrodermal Activity(EDA)

The electrodermal activity measures changes in the skin's electrical properties in response to autonomic signals. This correlates positively with mental workload variations (Wilson, 2001; Chapon & Gabaude, 2009). Since it is not very selective and can be impacted by numerous factors, including respiration, temperature, humidity, and arousal (de Warrd, 1996), it is commonly used as an indirect measure of cognitive effort (Critchley et al., 2000).

1.2.4 The ElectroOculoGram(EOG)

The electroOculograms record the activity of the eyes. Saccadic responses and peaks of saccadic velocity increase in complex situations, indicating a high mental workload. According to some, eye activity is more influenced by visual rather than cognitive factors.

2. Driving Behavior Studies using Driving Simulators

As the thesis has stated, research of this type should be conducted primarily with driving simulators, and some claim that only driving simulators should be used (Brookhuis & Waard, 2010). In general, behavioral research and laboratory methods rely heavily on simulators for their high precision, reliability, and controllability. Therefore, it is easy to change one or a few variables while keeping all others constant, making the driving simulator extremely

useful for comparative effectiveness studies. Keeping the environment and experiments constant ensures that all drivers are exposed to the same conditions, resulting in greater statistical power than in field studies (Nilsson, 1993). The driving workload experiments can pose many challenges, sometimes leading drivers to be "forced" into unwanted, more or less critical situations. It may be related to high-speed levels, complex workload scenarios, increasing variables for visual demands, and short time intervals for detecting deciding, and acting. Driver behavior under stressful conditions can be studied in the simulator, even in cases where a collision or a crash might occur (Petit et al., 1990; Haraldsson et al., 1991). In addition, it is possible to monitor driver actions before critical situations and accidents occur. By monitoring driver behavior and problems, research can be conducted to answer questions like: When do drivers need support? How can they be supported? Furthermore, how should a support system be designed?

Several studies are available in the literature concerning the validation of driving simulators (Tornos, 1998; Godley et at., 2002; Yan et al., 2008). It is almost impossible to predict what will happen in real traffic in every situation since they vary in every respect. There are also "unsubstantiated" events with varying degrees of uncertainty. No matter how advanced the simulator, it will never be able to reproduce the described reality accurately. In some cases, non-complex driving tasks and tasks that only require

a few details and components can be studied very well with a complete representation of driver behavior. In order to take advantage of methodological possibilities and avoid drawback effects, a behavioral researcher must determine which studies to conduct in which environment. The combination of field and simulator studies is powerful and valuable when appropriately applied. Mismatches between movements and visual impressions usually cause motion sickness. Subjects' tendency to experience motion sickness varies, so it is impossible to predict who will be sick in a driving simulator. Based on Neilson (1993), having a screen in front of you can contribute to impaired well-being in some drivers. The strongest indication of motion sickness in a simulator is a driving task requiring a lot of fast and large maneuvers and many sharp bends.

3. Gaze Behavior

A human's natural and most repetitive motion is shifting their gaze from one point to the next. This activity occurs automatically when we watch a video or read a book. Involuntary eye movements represent the greater portion of eye movements. They are efficient and frequently used. As a result of the massive amount of visual information surrounding us, we are required to select specific portions of our visual field to better process stimuli. The physiology of the retina requires the eyes to move constantly in order to acquire an accurate image. The frequent movement of the eye is referred to as saccadic movement. For a short period during the exploration of an image or scene, the eye tends to fixate on the most significant areas for a few milliseconds before moving to a new area. A human eye's trajectory consists of fixation periods interrupted by saccades, which move the eyes from one region to another. Fixation periods are when the eye is almost stationary, which allows the brain to process data. The duration of these periods is between 70ms and several hundred seconds. Second, in higher-level analyses, smaller eye movements, such as tremors, drifts, and flicks, are generally not significant at higher levels of analysis.

However, there is yet to be a widely accepted or well-defined method for performing this analysis. Some researchers analyze gaze data by examining saccades rather than fixations (Peters and Itti, 2006; Carmi and Itti, 2006). A saccade is a rapid movement of

the eyes aimed at repositioning a new region on the retina. This saccade detector will identify saccades using a velocity threshold, and a method also used to detect fixations (Fixations fall below the threshold, and saccades fall above it).

3.1 Fixation as a workload indicator

To provide an introduction to fixation, the thesis overview various attempts of research related to fixation. In Jacob & Karn (2003), eye fixation refers to a relatively stable eye position within a dispersion threshold of 2 degrees over a minimum duration of 200ms at a velocity threshold of 15-100 degrees per second. Eye fixation measures have been used in several studies. Efficiency negatively correlates with fixation number (Goldberg & Kotval, 1998), while efficiency negatively correlates with workload. A related metric, fixation frequency, correlates positively with the subject workload, as there is a strong correlation between fixation duration and the test run duration. Normalizing the metric is required to apply it across subjects. According to Van Orden, Limbert, Makeig, & Jung (2001), fixation frequency indicates tremendous effort in memory tasks, resulting in poorer performance accuracy and longer search times. Based on the previous analysis, the variance of fixation duration is determined by comparing its frequency to the total data collection. The longer the fixations, the greater the cognitive load (Callan, 1998).

It must be noted, however, that different studies employ different algorithms and settings for identifying fixations and saccades from

eye-movement records and for filtering out bad samples, blinks, and other anomalies. Though these variations in algorithms and settings have contributed to a consistent overall finding among eye movement studies, previous research has also found differences between individuals, stimuli, tasks, and eye movements, raising the question of whether thresholds for distinguishing fixations and saccades should also vary.

3.2 Identification Algorithms

Identification is often a crucial aspect of eye-movement data analysis that can significantly affect later analyses. A poorly defined identification algorithm may result in too many or too few fixations or may be overly sensitive to outlier data points, thus biasing interpretations. Karsh and Breitenbach (1983) demonstrated how different identification algorithms could produce drastically different interpretations despite analyzing identical protocols. Consequently, good fixation and saccade locations and durations are assured with good identification algorithms, which are then used to make inferences about processing complexity and visual demand. The following descriptions of each algorithm were addressed from Salvucci and Goldberg's research on eye-tracking protocols (2000).

3.2.1 Velocity-Based Algorithms

Using velocity-based algorithms, eye-tracking protocols utilize the fact that fixation points have low velocities and saccade points

have high velocities to discriminate fixation and saccade points. One of the easiest identification methods to understand and implement is Velocity-Threshold fixation identification (I-VT). Based on point-to-point velocities, I-VT separates fixation and saccade points (Erkelens & Vogels, 1995; Sen& Megaw, 1984). Velocity profiles of saccadic eye movements reveal two distributions: low velocities for fixations and high velocities for saccades. Velocity-based discrimination is rather straightforward and robust due to saccadic eye movements. Although it is possible to create a locally adaptive velocity-based fixation identification algorithm, velocity profiles have strong physiological and physical underpinnings, making static criteria adequate.

Hidden Markov model fixation identification (I-HMM) uses probabilistic analysis to determine the most likely identifications for a given protocol. Using a two-state HMM, HMM decodes velocities to identify fixation points or saccade points, collapses fixation points in fixation groups, removes saccade points, and maps each fixation group to the center of its points.

3.2.2 Dispersion-Based Algorithms

Unlike velocity-based identification of I-VT and I-HMM, Dispersion-Threshold Identification (I-DT) exploits the fact that low-velocity fixation points tend to cluster close together. Under the assumption that fixation points generally occur near one another, dispersion-based algorithms emphasize the dispersion of fixation points. In I-DT, fixations are a collection of consecutive points within a specified dispersion or maximum separation (Widdel, 1984). Since fixation typically lasts at least 100ms, dispersion-based identification techniques often implement a minimum duration threshold of 100-200ms (Widdel, 1984) to minimize equipment variability. This I-DT algorithm is based on Widdel's (1984) data reduction algorithm. Two parameters are required for the I-DT algorithm: the dispersion threshold and the duration threshold. Using exploratory analysis of the data, it is possible to estimate the dispersion threshold. According to task processing demands, the duration threshold is usually set between 100 and 200ms.

MST identification (I-MST) involves connecting a set of points to form a tree whose line segments have a minimum length. For dispersion-based fixation identification, MSTs can provide a highly flexible and controllable representation (Goldberg et al., 1995). The MST must first be constructed, and then it must be searched. The construction process uses prim's algorithm (Camerini et al., 1988). There is only one MST for each set of points. With MST

representations, dispersion analysis can be controlled, flexible, and locally adapted, and subsequent characterization of defined fixations can be improved.

3.2.3 Area-Based Algorithms

Area of Interest Fixation Identification (I-AOI, DenBuurman et al., 1981) identifies fixations within specific target areas. Described as rectangular areas of interest, target areas are units of visual information. Target areas retain identified fixations close to relevant targets during later analyses, such as tracing. In addition, I-AOI uses a duration threshold to distinguish fixations in target areas from saccades passing through.

When combined with I-AOI, a large picture can be obtained than when fixations are used alone. Like a fixation, dwell time within an area begins and ends at a specific time. AOI is determined by the data set of fixations, so the time between dwells does not describe saccadic behavior. A variety of saccades and fixations could occur between AOIs. Even though AOI-based dwell-time algorithms are not fixation algorithms per second, they are valuable as a concept to explain higher-level collections of fixations organized around a visual target or area. Table 3 shows a description of representative algorithms for two criteria.

Table 3. Description of Representative algorithms regarding two criterias

Criteria		Representative Algorithms				
		I-VT	I-HMM	I-DT	I-MST	I-AOI
Spatial	Velocity-based Dispersion-based Area-based	○	○	○	○	○
Temporal	Duration sensitive Locally adaptive		○	○ ○	○	○

3.3 Fixation while Driving

Research advocates that assessing gaze behavior is closely related to visual demand while driving. The theory may suggest that research must look at Visual functional-field size and fixation duration during the task. Drivers tend to concentrate more on the vehicle's front as cognitive demands increase. As a result, it has been associated with reduced perception of peripheral and central targets and reduced glances at mirrors and speedometers. When the driver shifts glance away from the primary driving task for long periods, he or she becomes less aware of the road and other road users nearby.

Recarte et al., (2000) researched the impact of verbal and spatial-imagery tasks on eye fixations while driving. Twelve participants performed two verbal and two spatial-imagery tasks on each route. During the spatial-imagery task, participants fixed their eyes for longer than during ordinary driving. As a result of the spatial-imagery task, driving performance decreased with a decrease in glances at mirrors and speedometers. Results are interpreted according to multiple attention-resource theories; internal distractions are discussed as impaired information

processing affecting road safety.

A study conducted by Wierwille found that these glances last between 500 and 1600 milliseconds (1993). In order to perceive information related to both driving and non-driving tasks, drivers simultaneously glance between primary and secondary tasks. Distractions inside or outside the vehicle might divert the driver's attention from the road and compromise the driver's ability to drive. Current trends suggest that road traffic injuries will become the fifth leading cause of death by 2030 (World Health Organization., 2013). Inattention among drivers is a major factor in road accidents. A study by the NHTSA found that distraction or inattention contributes directly or indirectly to 16% to 80% of traffic accidents. In the context of drivers, distraction is commonly defined as "diverting attention from critical activities for safe driving to competing activities" (Regan et al., 2008). The two main components of this condition are cognitive and visual distractions (Victor, 2005). It has been widely shown that these activities affect driving in several ways (Langle et al., 1999; Ranney et al., 2000; Muhrer & Vollrath, 2011; Kountouriotis et al., 2015; Kountouriotis and Merat, 2016). The consequences and causes of visual distraction on Lateral driving performance are relatively straightforward. Lateral control is impaired by visual distraction in comparison to baseline conditions (Kountouriotis & Merat, 2016), resulting in a significant increase in lateral position standard deviation (SDLP, Kountouriotis and Merat, 2016), a significant risk

of land departure, and a shorter crossing time (Engstrom et al., 2005). It is believed that this occurs due to increased eye glances off-road during the completion of visually distracting tasks. This part will overview Spatial Gaze Distribution, pupillometry, and blinks to introduce a methodology developed as a formal framework. The following review intends to support future research, which seeks a mythological approach to gaze behavioral data analysis to examine visual demand while driving.

3.4 Interpretation of Spatial Gaze Concentration

Over the past several years, eye gaze data have been used to characterize changes in cognitive load while driving (Harbluk, Noy & Eizenman, 2002; Sodhi et al., 2002; Victor et al., 2005). Percent road center (PRC) is one measure that is intuitive for measuring changes in drivers' gaze under added cognitive demands. A PRC is defined as the percentage of fixations that fall within a predetermined road center area over time. It has been shown that PRC increases with heightened cognitive demands (Engstrom, Johansson, & Ostlund, 2005; Harbluk et al., 2002, 2007; Recarte&Nunes, 2000; Victor et al., 2005). The PRC method uses a rectangle 15 degrees wide, directly in front of the vehicle. An analysis presented by Victor et al., (2005) looked at a rectangular region of 20 degrees. Another study by Victor et al., (2005) defined the road center as a 16-degree circle centered around the road center. Further developing this methodology, Ahlstrom,

Kircher, and Kircher (2009) examined a circular region; also 16 degrees in diameter, centered on the driver's most frequent gaze angle. It is conceptually easy to understand PRC, but individual studies use significantly different definitions of road centers.

PRC based on central gaze defines the road center based on the participant's most frequent gaze angle rather than world coordinates (Ahlstrom et al., 2009). For all highway driving periods of the session (including non-experimental measurement periods, which occurred about twenty minutes before the secondary demand), gaze angle data were binned into two-dimensional histograms extending horizontally and vertically by 50 degrees. The bin with the most frequent gaze angle was selected as the gaze center. If multiple angles were equally frequent, the one closest to 0 was chosen. Using the gaze center as a starting point, a circular area was defined as the road center. Alstrom specified an 8-degree radius for road centers but recommended examining other radii.

4. Pupillometry

In recent decades, it has been established that changes in pupil diameter are indicators of cognitive activity. German neurosurgeon Bumke had already recognized at the beginning of the previous century that intelligence or physical activity results in pupil enlargement (Hess 1975). An eye exam for patients that includes pupillometry evaluates the pupils and reactivity of the pupils, which

is an essential part of the clinical neurological exam. Increasing pupillary responses can indicate that cognitive activities are occupying the brain. According to Granholm and Steinhauer (2004), pupil dilation is associated with increased brain processing.

4.1 Measuring Pupil Dilation

Two groups of muscles within the iris are primarily responsible for pupil dilation. The pupil has a set of muscles encircling it; when activated, these muscles narrow the pupil's diameter. An additional set of muscles extends directly from the pupil through the iris, immediately outside the circular muscles. Activated radial muscles increase pupil diameter by pulling it outward. Both agonistic and antagonistic responses are typically involved in this reciprocal innervation process.

Video-based eye trackers use cameras and infrared illuminators to capture eye movements and corneal reflections. The cameras are placed either in front of the computer screen the subjects are viewing (desk-mounting) or on a helmet that places them directly in front of their eyes (head-mounting). As pupil images are recorded, the eye tracker can measure pupillary dilation by counting the pixels in the pupillary area or by fitting an ellipse to the image and calculating its major axis length. (Granholm& Steinhauer, 2004)

By measuring the movements of subjects' pupils relative to the cornea's reflection, the eye-tracker can track their eye movement.

During the calibration procedure, the participants are shown black dot targets, and they are instructed to gaze at them. Tracing sensors on the edges of the screen allows for further adjustments based on head movements if the eye tracker is head-mounted.

4.2 Interpretation of Pupil Response

Memory load, cognitive difficulty, valence, arousal, pain, and many other factors can cause pupil dilation in humans (Beatty, 1982). According to Peavler (1972), for individual responses, dilation would take place after emotional stimuli were presented and at a faster rate for stronger stimuli. As demonstrated by Partala and Surakka (2003), subjects began to respond to different sounds (baby crying, laughing, and regular office noise) quickly after stimulus onset, with pupil sizes increasing steeply after 400ms. It has been shown that pupils dilate when confronted with cognitively demanding processes, peak about one to two seconds after the onset of demand (Beatty, 1982), and constrict once the task has been completed, either gradually (Hess, 1975; Kahneman & Beatty, 1966) or instantly (Bernhardt, Dabbs, & Riad, 1996).

A review of the relevant literature by Beatty (1982) concluded that task-evoked pupillary responses were reliable indicators of cognitive load within tasks, across tasks, and between individuals. To report on each time interval of interest, Hense, Beatty, and Lucero Wagoner (2000) suggested three measures: mean pupil dilation, peak dilation, and latency to the peak. A baseline was

established, and researchers compared these measures against it. As noted by Beatty and Lucero-Wagoner (2000), percentage measures are inflated when the pupil size is small at baseline, so absolute measures are recommended.

4.3 Pupil Dilation as a workload Indicator

It has been observed that pupil diameter increases with increasing cognitive load across multiple tasks and conditions (Alneas et al., 2014; Chen & Epps, 2014; G. Brown et al., 1999; Klingner et al., 2011). Kramer (1990) describes the effect of cognitive load on parasympathetic activity in the peripheral nervous system, which increases pupil size. It has been demonstrated that this type of behavior manifests in various tasks, including short-term memory, language processing, reasoning, perception, sustained attention, and selective attention, according to Beatty (1982). Additionally, it satisfies Kahneman's three criteria for determining task difficulty: reflecting differences in task difficulty within a task, between different tasks, and between different individuals (Kahneman, 1973).

Researchers (Redlich, 1908; Westphal, 1907) found a link between physical task demands and pupil dilation. A change in pupil dilation is also correlated with a change in the viewing angles of the photograph (Hess, 1975). In recent years, researchers have developed a metric for measuring pupil dilation frequency and power to detect cognitive load (Gavas, 2017; Duchowski, 2018).

Pupil dilation was recorded during several driving studies when secondary cognitive tasks were used to induce mental workload (Recarte& Nunes, 2000; Wierwille et al., 1994). A consistent increase in pupil diameter was observed when performing spatial imagery and secondary verbal tasks. Palinko et al., (2010) examined mental workload in simulated driving environments using a remote eye tracker. Also, Palinko et al. introduced a new approach to measuring pupil diameter, called the mean pupil diameter change rate (MPDCR). According to these authors, this new measure is beneficial for monitoring rapid changes in mental workload.

Despite this, there have been some limitations in hardware and software that make measuring pupil changes over time on complex tasks difficult. It only took until recently for a technique to be developed that allowed reliable and rapid estimation of cognitive workloads from pupil dilation changes to be developed. Index of Cognitive Activity (ICA) is the technique used.

4.4 Index of Cognitive Activity(ICA)

In summary, pupils that dilate more during task completion are associated with more significant cognitive load. However, pupil size can be misleading as a measure of cognitive load, as pupils dilate and contract based on how much light they receive. Therefore, the brightness of the visual scene in the research environment should be carefully controlled to prevent confounding between workload and light reflex dilations. In addition, pupil dilation and contraction are too slow to accurately depict mental responses to stimuli that follow each other rapidly or overlap. This may pose a problem for tasks such as driving, which requires continuous steering movements (Demberg et al., 2013), traffic sign comprehension, where words are presented frequently (e.g., Hyönä et al., 1995), and in-vehicle tasks (Demberg et al,. 2013). A different pupillometric measure of cognitive load, the Index of Cognitive Activity (ICA; Marshall, 2002), has been developed to overcome these disadvantages; however, it has received very little attention until recently. The ICA is computed by assessing the number of rapid increases in pupil size over a specified period(Marshall, 2002; Demberg, 2013). By distinguishing high and low-frequency properties from the signal, it separates rapid dilation changes resulting from a cognitive activity from slow ones resulting from variations in light input.

Additionally, the ICA has a short latency relative to pupil size and low auto-correlation. A low auto-correlation means that the

frequency of rapid dilations in a certain time frame shows very little correlation with the frequency of rapid dilations in a previous time frame (Demberg and Sayeed, 2016). The ICA has a short latency and a low autocorrelation, making it ideal for experiments in which high time resolution is crucial for determining the relationship between the measured effect and the stimulus (Demberg and Sayeed, 2016). It is possible to obtain the ICA using a conventional eye-tracking device (remote), so it can also be used to assess cognitive load in natural environments, such as driving, more easily than other more fine-grained, but more cumbersome, physiological measurements, such as EEG, heart rate, or skin conductance. Due to the existing evidence, the ICA can be used reliably to measure cognitive load and is complementary to the conventional pupil size measure, especially when analyzing finer details under conditions where constant lighting is tricky. However, it is still a challenge whether the ICA has the same cognitive underpinnings as pupil size as a whole or is influenced differently. Especially there needs to be more information about the ICA's response when multiple tasks are being performed concurrently. Thus, the primary objective is to systematically examine ICA as a sensitive measure of processing vision-workload due to traffic environment. The aforementioned ICA overview was described more in detail by Vogel et al., (2018) in “The Index of Cognitive Activity as a measure of cognitive processing load in dual-task settings”. If you would like a more detailed explanation

of how the index is calculated from raw pupil data, please refer to Marshall (2002). It has been shown that the Index of Cognitive Activity is sensitive to various levels of mental load for different cognitive tasks. A higher ICA value was observed in scenarios or tasks requiring high cognitive processing levels. Therefore, the index provides a valuable measure of cognitive activity.

5. Blink

It has been investigated in numerous studies whether blinks provide complementary information about the processing of information over time. In independent literature, blinks are not a random phenomenon (Stern, Walrath, & Goldstein, 1984) but instead, occur in response to high cognitive load (Fukuda, 1994, 2001) or information processing (Ichikawa & Ohira, 2004), possibly reflecting the release of cognitive resources associated with stimulus-related cognition (Ohira, Winton, &Oyama, 1998). As a result of preparation, eyeblinks have also been observed to be suppressed directly preceding cued information processing (Ohira, 1995). It is also possible to blink just after preparation since blink latencies of very short magnitude are associated with errors in cognitive tasks when there is insufficient preparation (Sirevaag et al., 1999). According to Marx et al., (2003), cognitive modulation of blinks may be associated with transitions between interoceptive states associated with closed eyes and exteroceptive states associated with open eyes.

5.1 Interpretation of Blink

Stern et al., (1994) demonstrated that blink could be affected by the perceptual demand of tasks and cognitive variables: experimental evidence was recently provided by Recarte et al., (2008), which came to the conclusion that Blink rate is affected by both mental workload and visual demand, which act in opposition

to each other, the former leading to blink rate increase, the latter to blink rate decrease. Through a aforementioned review, the thesis concludes that blink data must be carefully analyzed during the process.

6. Conclusions from review

During the past decade, workload research has been integrated into existing stress assessment research, and eye-tracking technology has spurred the investigation of this phenomenon. Galinsky, Rosa, Warm, & Dember (1993) found that participants frequently reported increased stress levels after completing workload tasks, suggesting that additional factors may contribute to the perception of workload. Even though vision is considered the most important resource available to human, most workload research did not use visual physiological data to examine how drivers initiate changes, how the task at hand is performed, how the task environment is set up, or a combination of those variables. Changing the psychophysical task parameters in workload studies moderated workload but did not significantly decrease stress measures.

As a result of the literature review, many descriptive workload and performance models have been proposed that, during driving, a driver's workload is increased by additional task demands: if the workload is low, drivers can cope with it by using various coping strategies. A driver's performance, however, declines when his task demands exceed a certain level. It is still being determined which

quantitative changes are caused by an increase in workload or a reduction in performance with these models due to the lack of a recognized, sensitive, instantaneous, and non-invasive method of assessing workload. Furthermore, traditional measures need to reveal how drivers' coping strategies affect driving performance concerning workload. Because these coping strategies vary from person to person, and some strategies are more effective than others, they all affect the ability to remain focused and drive safely. Due to this, existing models of workload and driving performance are too simplistic, and they are not suitable for transportation applications.

This review introduces how research in vision studies cognitive load through gaze behavior, pupillometry, and blinks. A description of how vision researchers have used velocity and dispersion to categorize fixation and saccadic movement was provided. It has been suggested that the pupil dilates and blinks eyes during cognitive effort and the degree of dilation and blinking depends on the difficulty level. Additionally, various video-based eye trackers using cameras and infrared illuminators were discussed.

As previous traffic research regarding workload and safety did not look at visual physiological factors, the thesis investigated recent attempts in which these measures were integrated as dependent variables. Following that, the thesis will examine visual measures with the purpose of developing them in light of complex case studies.

Chapter 3 Methodology

1. Development of Driving Vision-Workload concept

Our approach to Driving Vision-Workload was based on reviewing previous studies that evaluated workload, visual tools used in background knowledge studies, and driving experiments used in observable behavior studies. Physiological measurements, subjective reports, and the experimenter's performance have been used in existing studies to verify workload. Several versions of emotional, physiological, and visual-behavioral measures were identified during the review process. Moreover, similar scales were used in prior driving experiments to analyze drivers, and the results were confirmed by analyzing drivers through related scales. Accordingly, the thesis aims to reflect and define the workload generated during driving and the visual characteristics derived from previous studies by applying the Driving Vision-Workload concept.

The finding conclude that the amount of work required to visually collect information while driving is known as driving vision-workload, and it presents a danger to traffic safety. Behaviors associated with driving include lane keeping, reaction time, blinking rate, pupil behavior, and gaze point of the driver.

2. Experiment Design

2.1 Apparatus

In this experiment, the variables of the fixed-base driving

simulator used were acceleration, deceleration, and lateral position. Regarding pedal use, the intensity of pressing the gas and brake pedals ranges from 0 to 1. For instance, "0" denoted that the deal was not pressed, while "1" denoted that the pedal was pressed to its full depth. The driving speed and gas pedal power were collected every 10m before the sign, and the values were averaged for all participants.

2.1.1 Eye Tracker

In the experiment setup, Tobii Pro Glasses 3, a head-worn eye tracker connected to a recording device that held the saved data from the eye tracker, was used. This is a binocular eye-tracking setup with two cameras and six glints per eye, and it was chosen because it is the most accurate eye-tracking unit available.

Using an ethernet cable, the Tobii Glasses recording device was connected to a desktop running the Tobii Glasses Controller software (version 1.11.4) (Tobii Technology, Stockholm, Sweden). The system was calibrated using the one-point calibration procedure provided with the eye-tracking setup in 50Hz mode. The calibration consisted of (a) Fixing the marker to the center of the stimulus grid, (b) instructing the participant to fixate on the marker's center, and (c) entering the calibration mode in the Tobii software (Tobii Technology, Stockholm, Sweden), which then completed the calibration process automatically. The front-facing cameras recorded a video stream at 25 Hz with a resolution of

1920 X1080 px, and the four-eye cameras recorded a stream at 50Hz containing Four eye images(Two views of each eye) with a combined resolution of 240 X 960 px. Using the Tobii Manual definition of fixation, the study defined fixation as the interval between the end of one saccade and the beginning of the next saccade. A description of an employee's eye tracker can be found in Figure 2.



Figure 2. Tobii Pro Glasses 3

2.1.2 Driving Simulator

A virtual reality modeling program, UC-win/Road, was first used to simulate construction, urban planning, civil engineering, and traffic. By combining the program with a driving simulator, several driving situations in diverse environments can be emulated. An automatic car was utilized in this study, which included a seat, a seat belt, a steering wheel, a speedometer, an ignition, and key system, an accelerator and a brake pedal, as well as an ignition and key unit. Figure 1 illustrates the visual display system, which consists of three color monitors with a 180-degree field of view synchronized to display a realistic road environment generated by a computer. Additionally, the simulator included an audio system

that provided realistic traffic sounds and instructions. Figure 3 shows a description of the driving simulator. Driving experiment was conducted under gloomy weather conditions in daylight with two main sections, tangential and curved.



(a)



(b)

Figure 3. Location of study sites

(a) Driving Simulator; (b) Driving Simulator in progress of the experiment

2.1.3 Scenarios

The simulations depicted a virtual-reality reconstruction of a two-lane rural road in Gangwon-do province, with horizontal curves and tangent segments. The highway segment was a 5.56 km section of South Korea Route 28. According to the topographic layout, a straight road was the basis for this reconstruction. As a result, the simulation was designed to provide a similar road perception by incorporating the 3.6 m lane width and 140 m curve radius of the roadway segment, as well as other road engineering characteristics. Several scenarios with several combinations of experimental conditions were designed and implemented in virtual reality. In the experiments, two geometric conditions were used, including a curved and tangent segment, and two warning signs were used for black ice warnings. A random order of collaboration was used for each case. The signs were located 22 meters after the end of the curve segment and 200 meters after the end of the Tangent segment. Yellow and white road markings were used with a 10 cm width. During the straight segment, signs influenced 200 meters of the road. Signs influenced approximately 75 meters of the curved segment's road length. It is illustrated in Figure 4 that two typical sign locations, i.e., a simple design sign and a complex design sign, were considered in the four experiments involving signs. Figure 5 illustrates how the ends of curve sections and points in tangent sections were taken into account to analyze their effects on driving behavior.



(a)



(b)

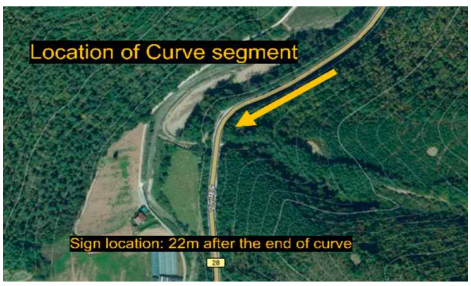


(c)

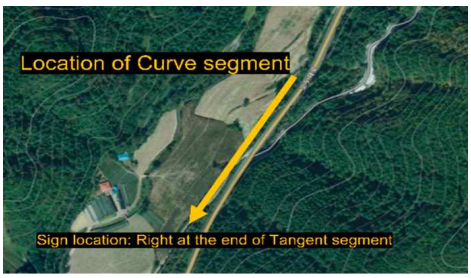


(d)

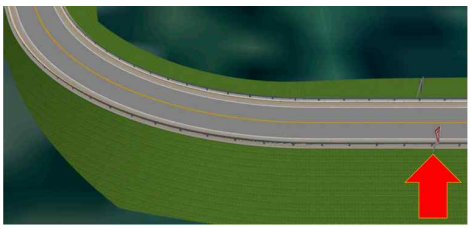
Figure 4. The signs used in experiment: (a) simple design black ice sign; (b) complex design black ice sign; (c) simple design black ice sign in experiment; (d) complex design black ice sign in experiment



(a)



(b)



(c)



(d)

Figure 5. Location of study sites and the signs on the segments for experiments: (a) curve segment study site; (b) tangent segment study site; (c) curve segment in experiments; (d) tangent segment in experiment

2.1.4 Procedure and Participants

In the experiment, each participant was tested individually. Upon arrival, the participants were briefed on the requirements of the experiment. As part of the informed consent process, all participants read and signed an informed consent document and completed a short questionnaire (e.g., age, gender, etc.).

Afterward, the driving simulator was introduced, and its features were explained to the participants. As they were familiarized with the vehicle in a neutral state, they were given the opportunity to drive it. During this condition, participants reconstructed scenario 1 in reverse, familiarizing them with the torque-based tactile feedback offered by the steering wheel and pedals.

A mountainous road scenario was presented first, followed by underpass road scenarios. The participants were requested to adapt their speed to the road conditions as follows: "Drive as you would in the real world until you are instructed to stop. The speed limit on this road is 60km/h."

A questionnaire regarding hazard perception was given to participants after the fourth run. In this study, 37 healthy subjects were recruited from our panel to participate in a driving simulator experiment involving the visual perception of traffic scenes. Table 4 shows a description of the participants. The driving phase of the experiment lasted an average of 30 minutes. They took an average of 50 minutes to complete the experiments. Those who participated in the study had valid driver's licenses and normal or

corrected-to-normal vision (contact lenses or glasses). Those who showed signs of simulator sickness were excluded from the study. All participants reported driving daily for at least three years and holding a driver's license.

Table 4. Participant gender by age groups

Age	20s	30s	40s	50s	Total
Male	4	13	6	6	29
Female	2	2	3	1	8
Total	6	15	9	7	37

3. Data Analysis

3.1 Method to decide Measures

Analyses were conducted on each measure based on its sensitivity, diagnostic ability, validity, and accessibility. We reviewed the sensitivity of each measure, based on findings from the previous study, and found that it was highly responsive to dual-task experiments and environmental changes while driving. Through statistical significance analysis, each measure's validity was evaluated. Accessibility was evaluated based on the convenience of collecting data. Several accessibility indicators were categorized as indicating functional availability in transportation applications.

3.2 Data Analysis Method

The analysis identified core areas where drivers look while

driving based on spatial gaze distribution and area of interest analysis. To assess a driver's attention distribution, gaze concentration, gaze density, and gaze duration were analyzed. Driving vision-workload was determined by analyzing physiological factors such as pupil size and cognitive activity. In this thesis, the number and duration of fixation visits between two roadway signs were determined according to literature recommendations. In the experiment, driving vision-workload from visual demand was examined in relation to optical behavior using simple and complex design signs.

In order to analyze the data, the experiment compared the number of fixation visits and the duration of fixation, demonstrating the increased visual demands on drivers to comprehend the surroundings. Spatial gaze concentration was determined by tracking the eye tracking locations of the driver gaze. Analysis provided insight into where gaze points were most prominent. Through pupillometry analysis, the study demonstrated which driving conditions influenced the driver's pupillometry behavior change. At least, pupil dilation was studied as a driver's mental workload. The thesis continues below with a detailed explanation of each analysis and interpretation.

Chapter 4 Development of Measures

1. Decision of Alteration Measures

We began by conducting a comprehensive literature search to identify relevant studies for this study. Keywords and search terms included workload, driving workload, cognitive workload, mental workload, physiological measurements, driving simulators, eye, pupil, blink, fixation, pupillary response, pupillometry, and eye tracking. As a result of the literature search, a comprehensive search was conducted for relevant references. After completing the major search, backward referencing was performed to determine if any related studies could be included. An investigation of each research was conducted based on a range of criteria for study inclusion after a comprehensive literature search was completed. It was required to quantify the independent variable of cognitive workload and the dependent variable of vision behavior measurement to meet the inclusion criteria. Using restricted criteria, driving vision-workload measurements candidate list was compiled.

2. Assessment Analysis

The literature review introduces how research in vision studies cognitive load through gaze behavior, pupillometry, and blinks. Researchers have discussed how velocity and dispersion can be used to categorize fixation and saccadic movement. According to

some, the pupil dilates and blinks during cognitive effort, and the amount of dilation and blinking depends on the difficulty level. Additionally, various video-based eye trackers with cameras and infrared illuminators were discussed. Because previous traffic research regarding workload and safety had not included visual physiological factors as dependent variables, the thesis investigated recent attempts to incorporate these factors. As a result of the thesis, visual measures were then examined to develop them based on complex applications.

Selected visual behavioral measures included fixation rate, duration, spatial gaze concentration, pupil size and ICA, blink rate, and duration, as Table 5 shows. Each measure was investigated on sensitivity, diagnosticity, validity, and accessibility.

In terms of sensitivity, each measure was reviewed from their previous study finding; instead, the measure was highly responsive towards dual-task experiments or changes in the environment while driving. Based on the review, fixation rate, pupil size, and blink rate were diagnosed as 'okay' regarding the plain quantitative relationship between measures and driving workload. Following that, the fixation duration, spatial gaze concentration, pupil's ICA, and blink duration were deemed to have a relatively robust qualitative correlation with the driving workload.

Based on diagnostic criteria, fixation rate, spatial gaze concentration, and pupil size were rated as 'okay' regarding a wide range of third variables, such as individual differences in

stress-coping behavior while driving. A strong connection between visual demand and measures was assessed as 'good,' including fixation duration, pupil's ICA, blink rate, and duration.

The analysis examined each measure's validity through a statistical significance perspective. Fixation rate, pupil size, blink rate, and duration were rated 'okay.' Regarding statistical findings, spatial gaze concentration, fixation duration, and ICA were considered 'good.'

The convenience of collecting data was examined as the last category of accessibility. The study classified the results of each measure for accessibility as indicating the presence or absence of a functional availability in transportation applications. For this analysis, quantitative measures such as fixation rate and pupil size were labeled as 'okay' for simplicity. Then, fixation duration, spatial gaze concentration, and ICA were categorized as 'good' regarding current eye-tracking technology. Due to the absence of a technological algorithm through the eye-tracker employed in the study, blink rate and duration were labeled as 'poor.' The description of reviewed visual indicators rated criteria is shown in Table 5.

Table 5. Description of reviewed visual indicators rated criterias

	Sensitivity	Diagnosticity	Validity	accessibility
Fixation rate	○	○	○	●
Fixation duration	●	●	●	○
Spatial gaze concentration	●	○	●	○
Pupil size	○	○	○	●
Pupil's ICA	●	●	●	○
Blink rate	○	●	○	_
Blink duration	●	●	○	_

'–'=poor '○'=okay '●'= good

Considering all circumstances, gazing and pupil dilation demonstrate high sensitivity, diagnostic accuracy, and effectiveness when compared to each other. Eye-blinking showed low experimental ease compared to sensitivity, diagnosis, and effectiveness. This diagnosis is based on the eye blink measurement member of the Tobii Pro Glasses 3 used here. Consequently, this thesis selects the driver's fixation and pupil dilation data as potential measures of driving vision-workload in the driving experiment.

3. Application

This paper was an exploratory study designed to develop general eye movement patterns while driving. Many studies have found that drivers generally look at the vehicle's frontal area as they spend most of their time moving forward.

To study a driver's visual behavior during the change within vision, it is necessary first to identify core areas where drivers should be looking. However, for the thesis, the analysis will be weighted more towards the visual demand that the traffic environment caused through spatial gaze distribution and area of interest analysis for two traffic signs.

For spatial gaze concentration, gaze density and gaze duration were used to examine where participants attempted to concentrate while driving. To identify mental workload that road sign may bring, pupil size and index of cognitive activity was considered to be measures of driving vision-workload. As literature review concluded, the study selected a number of fixation visits and fixation duration to differentiate vision workload between complex sign design and simple sign design. Next, to determine whether pupillometry measures and fixation indicators were valid within geographical workloads assessment, the thesis compared tangent and curve segment pupillometry measurements without signs.

We also compared a regular roadway exposed to surfaces with an underpass to determine the validity of pupillometry measures regarding mental workload regarding environmental surroundings. A

modular framework is generated based on multidimensional phenomenon theory for analyzing, defining, measuring, and applying vision workload in various contexts.

Since the thesis intends to introduce the framework of visual research method integration on driving research, statistical analysis regarding visual behavior performance was prevented. The thesis intended to present how visual research measures could apply in workload research assessment while driving.

3.1 Fixation and Spatial Gaze Concentration

3.1.1 complex design signs vs. simple design signs

3.1.1.1 Area of interest analysis on different design signs

First, area of interest analysis for different design signs was compared. Then, analysis of fixation visit and duration on curve and tangent road segments under the influence of each sign was compared. This study aims to find evidence that vision demand levels correlate with the number of fixation visits and the duration of fixation.

Based on Table 6, complex design signs resulted in almost twice as many fixation visits than simple design signs on the curve (simple design signs fixation visits=25, complex design signs fixation visits=62) and tangent segments (simple design signs fixation visits=65, complex design signs fixation visits=133). Based on the results, simple design signs showed the lowest fixation visits for curved segments (25 visits) and tangent segments (65 visits). For the complex design, 62 fixation visits were recorded for the curved segment, and 133 fixation visits were recorded for the tangent segment. Regardless of road geometry, gaze behavior indicates that complex signs have a tremendous increase in visual demand, as Table 6 shown.

The fixation duration, which is the duration in which a driver's gaze is fixed at one target, may also represent the higher visual demands for drivers' comprehension of each sign. As Table 6 shown, fixation duration for complex design signs was longest in

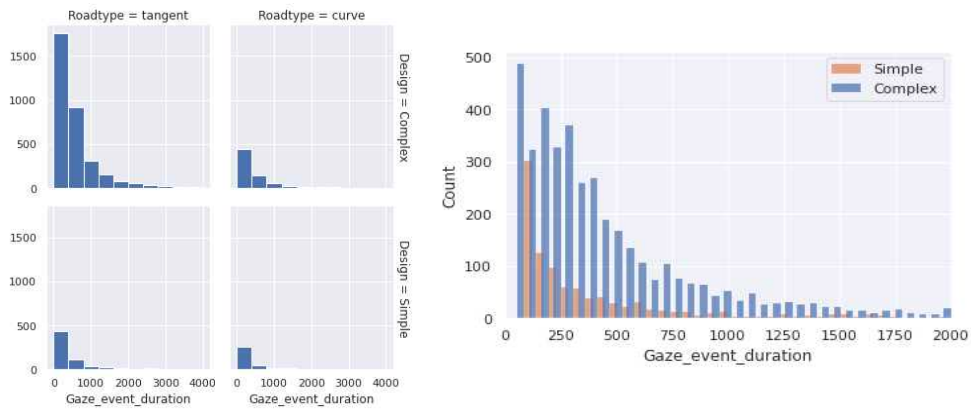
curved segments (M=352.03ms, SD=239ms) and tangent segments (M=606.67ms, SD=715.6ms). Meanwhile, simple design signs were 252.36ms (222.9mm) on the curve segment and 499,44ms (594.9ms) on the tangent segment.

Those who fixed their eyes on either sign longer than the fixation identification threshold, which was 60ms (Tobii Lab Attention velocity-based Algorithm) were recorded as those who fixed their eyes on signs. As the number shows, complex design signs brought 19% more people's attention to both curve and tangent segments.

Table 6. Area of interest analysis for each signs on two segments

Segment Type	Design of Sign	Number of participants who fixed eyes on signs(%)	Total Number of Fixation Visit (N)	Mean of Fixation Duration (Standard Deviation)(ms)
curve segment	simple	12 (32%)	25	252. 36 (222. 9)
	complex	19 (51%)	62	352. 03 (239. 0)
tangent segment	simple	16 (43%)	65	499. 44 (594. 9)
	complex	23 (62%)	133	606. 67 (715. 6)

According to Figure 6, complex design signs outnumbered fixation visit count for both tangent and curve segment.



Figures 6. Gaze duration multi-histogram for two signs(simple and complex design signs) on two road types(curve and segment)

3.1.1.2 Gaze Analysis on tangent segment for two signs

The distribution of attention for each geographical property was also examined using spatial gaze concentration analysis and area of interest analysis for two signs. Gaze density and gaze duration were used to examine where participants attempted to concentrate while driving. As for the tangent segment, complex design signs (3388 counts) showed 522% higher number of total fixation visits compared to simple design signs (648 counts) as well as 35% longer duration time (M= 613.42ms, SD=767.54ms, max=9137ms) as Table 7 shown.

Table 7. Gaze analysis of complex and complex signs on tangent segment

Type	MD(SD)	count	25%	75%	max	x(25~75%)		y(25~75%)	
complex	613.42 (767.54)	3388	210	711	9137	939	1050	461	613
simple	456.57 (598.76)	648	110	536	4769	923	1012	532	621

3.1.1.3 Spatial Gaze Concentration on tangent segment for two signs

For spatial gaze concentration through gaze density, complex design signs showed 24% horizontal, 70% vertically wider distribution than simple design signs, as Figure 7 shown on the tangent segment. Through gaze duration regarding spatial analysis, Figure 8 clearly showed a wider range of gaze distribution for complex design signs than simple signs. This would be an outcome influenced by a heavily concentrated gaze on the center, which narrowed the density graph in Figure 7. Figure 9 shows how fixation for complex design signs on the tangent segment is heavily concentrated in the center.

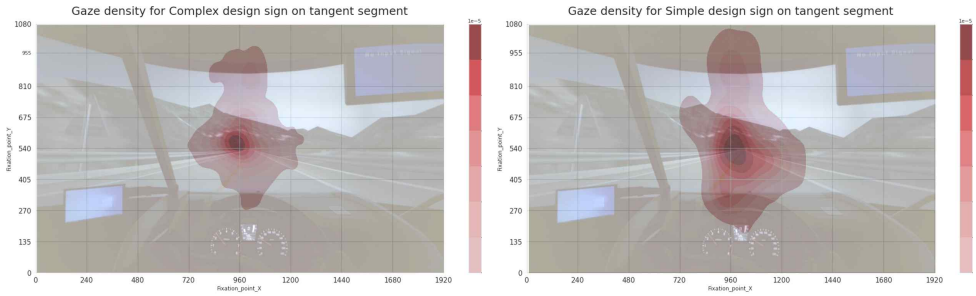


Figure 7. Gaze density for two signs on tangent segment

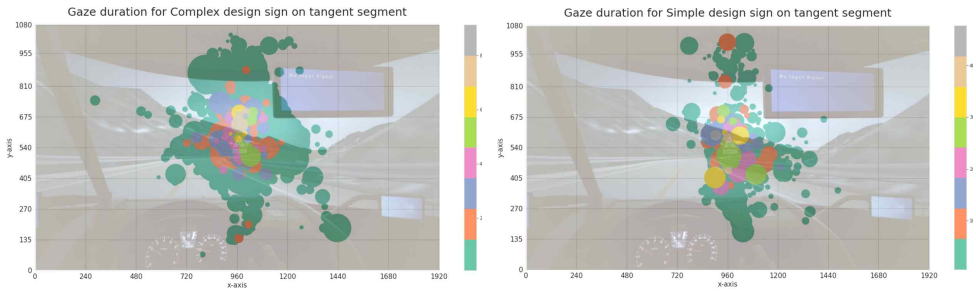
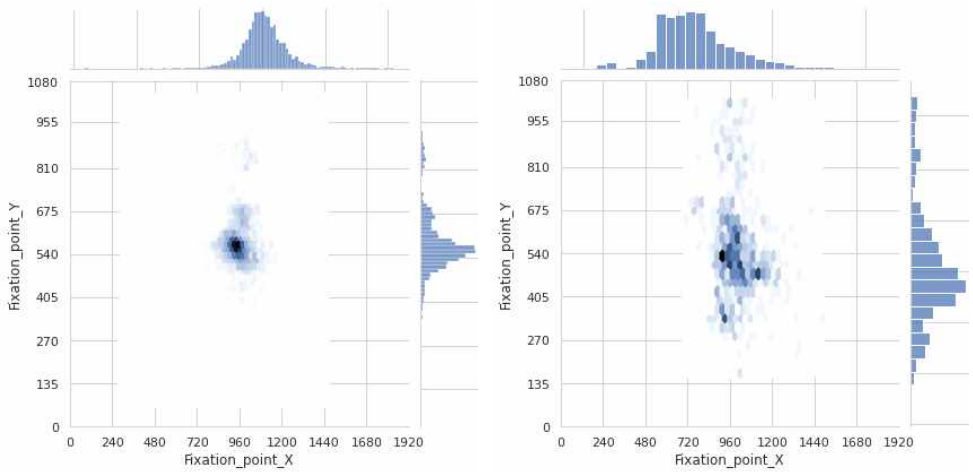


Figure 8. Gaze duration for two signs on tangent segment



(a) (b)

Figure 9. A jointplot for fixation distribution on tangent segment:

(a) complex design; (b) simple design;

3.1.1.4 Gaze Analysis on curve segment for two signs

For the curve segment, the complex design sign (697 counts) showed 205% higher total number of fixation visits compared to the simple design sign (339 counts) as well as 40% longer duration time (M= 493.36ms, SD=616.75, max=4959ms) as table 8 shown.

Table 8. Gaze analysis of complex and simple signs on curve segment

Type	MD(SD)	count	25%	75%	max	x(25~75%)		y(25~75%)	
complex	493.36 (616.75)	697	140	591	4959	875	994	490	606
simple	352.31 (476.59)	339	90	376	3146	852	951	363	589

3.1.1.5 Spatial Gaze Concentration on curve segment for two signs

The complex design showed 20% horizontally wider distribution for spatial gaze concentration. Meanwhile, the simple design showed 94% vertically wider distribution than the complex design.

Figure 10 and 11 show similar shapes of gaze dispersion across the participant's visual-field. Figure 12 shows similar horizontal eye movement distribution, while simple design signs showed more vertical dispersion of attention.

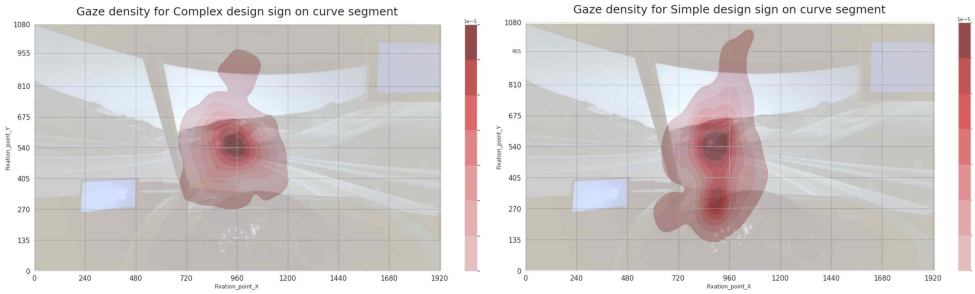


Figure 10. Gaze density for two signs on curve segment

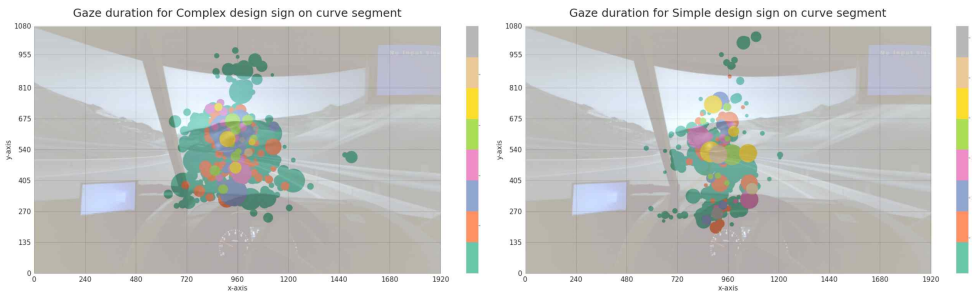
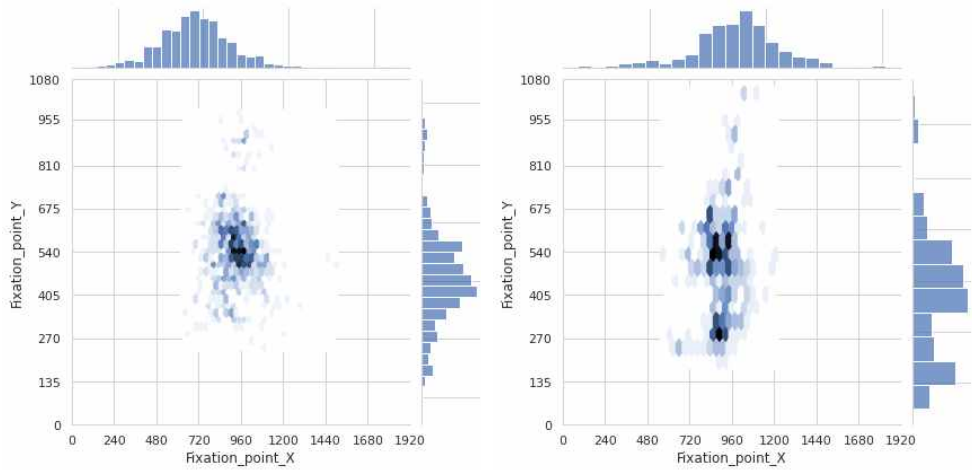


Figure 11. Gaze duration for two signs on curve segment



(a) (b)

Figure 12. A jointplot for fixation distribution on tangent segment: (a) complex design; (b) simple design.

3.1.2 Straight Road vs. Curve Road

3.1.2.1 Gaze behavior on geographical property

Next, the application investigated the properties of the driver's visual field regarding road types. Based on literature reviews, high visual and cognitive demand disrupt drivers' attention from the front of the vehicles. The hypothesis leans towards the change in cognitive demand for the road types that may influence gaze behavioral measures.

The tangent segment, which had the most extended max fixation duration (4047ms), showed some good behavior compared with the curve segment (2665ms) in free driving. The tangent segment showed 44% narrower concentration to the center horizontally and 1% wider vertically, accompanied by longer fixation duration (M= 298.99ms, SD= 304.11, max= 4047ms), as Table 9 shown.

Table 9. Gaze analysis of two types of segment without signs

Type	MD(SD)	count	25%	75%	max	x(25~75%)		y(25~75%)	
tangent	298.99 (304.11)	2727	110	371	4047	911	1016	497	601
curve	260.28 (240.72)	3333	100	331	2665	884	1041	496	598

3.1.2.2 Spatial Gaze Concentration on geographical property

As for the spatial gaze concentration, Both gaze density (Figures 13 and 15) and gaze duration (Figure 14) illustrated the apparent differences in gaze performance between tangent and curve segments. The tangent segment showed more spread gaze concentration vertically. Meanwhile, the curve segment showed

more concentration horizontally.

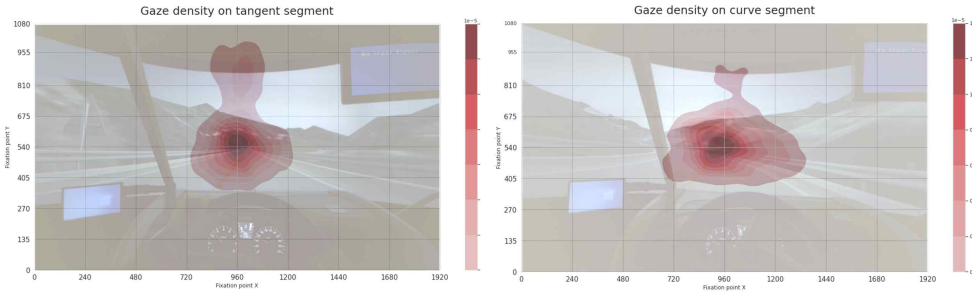


Figure 13. Gaze density for tangent and curve segment without signs

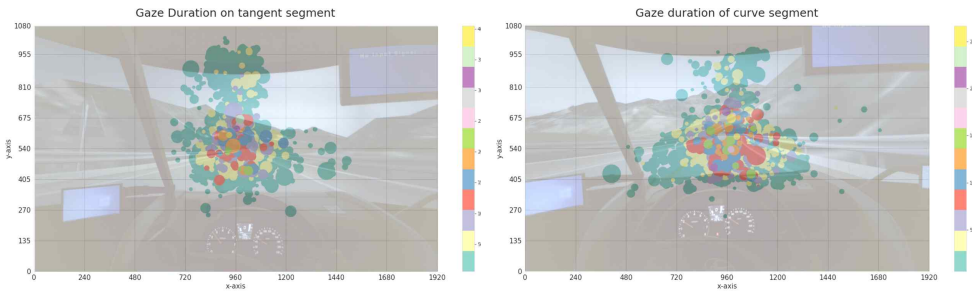
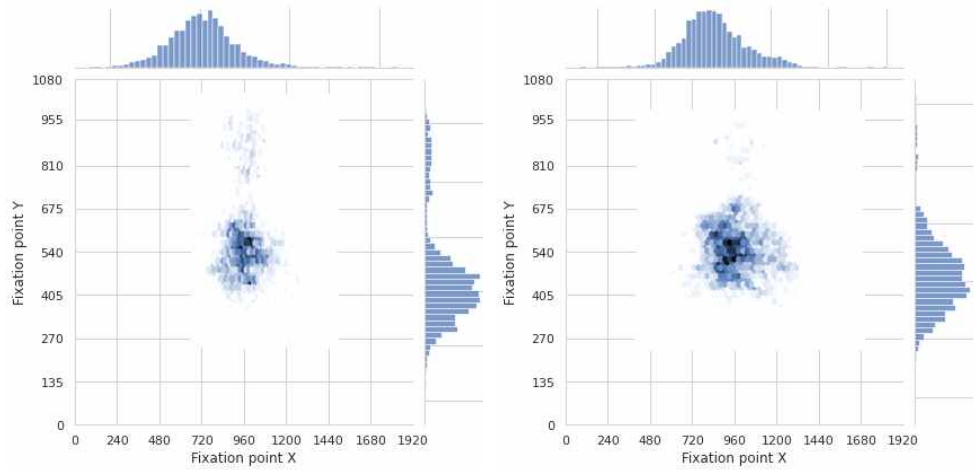


Figure 14. Gaze duration for tangent and curve segment without signs



(a) (b)

Figure 15. A jointplot for fixation distribution:

(a) tangent segment; (b) curve segment

3.1.3 Underpass

3.1.3.1 Gaze analysis on environmental property

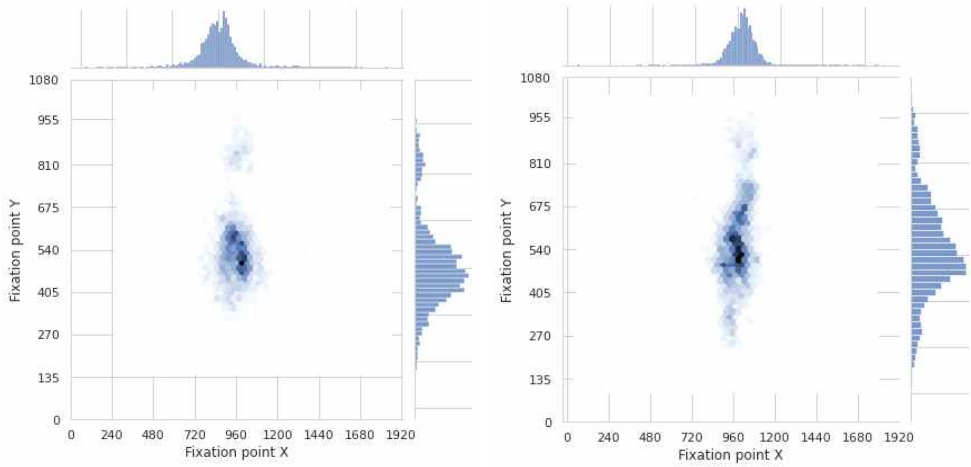
When performing driving tasks on underpass condition, the primary visual performances regarding gaze behavior did not change dramatically compared to above the ground condition as Table 10 shown. Majority of participants' number of fixation, gaze duration and spatial gaze concentration did not show an obvious difference. It was found that performing the driving tasks while driving in upderpass does not significantly change the number of fixation(5226 count), duration time(M=634.65ms, SD= 996.95ms, max=15598ms).

Table 10. Gaze analysis of for above and underpass

Type	MD(SD)	count	25%	75%	max	x(25~75%)		y(25~75%)	
above	577.35 (1094.58)	5765	140	624	48398	913	1015	471	596
underpass	634.65 (996.93)	5226	140	678	15598	940	1080	479	647

3.1.3.2 Spatial Gaze Concentration on environmental property

A steady consistency in gaze behavioral data across all levels was observed as Figure 16 and 17 indicated. Both above and underpass fixation dispersion were identically similar on both horizontal and vertical perspective. Figure 17 suggests that there is not significant difference in gaze duration analysis.



(a) (b)

Figure 16. A jointplot for fixation distribution:

(a) above;(b) underpass

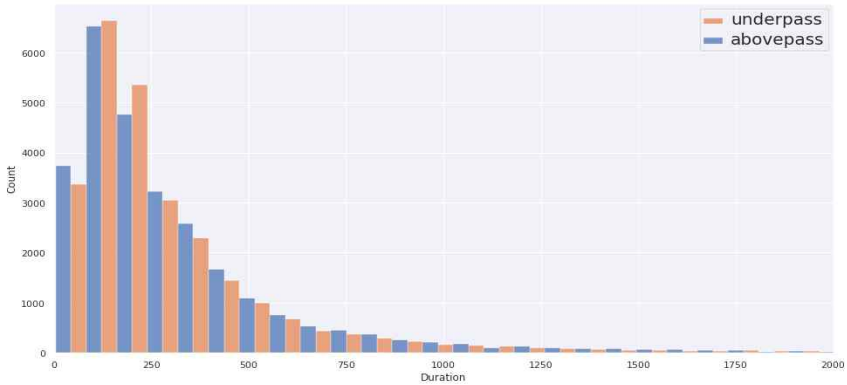


Figure 17. Gaze duration histogram for above and underpass

3.2 Pupillometry Analysis

3.2.1 simple design signs vs. complex design signs

3.2.1.1 Pupil dilation analysis on for two signs on tangent segment

the described pupillometry analysis were applied to the framework, which contains a pupil dilation size that indicate

driver's mental workload and also contains desciptions of ICA and further parameter of pupil size(25%~75%).

In terms of pupilometry comparison of two signs on tangent segment, complex design signs showed 11% increase in size(Mean=4.52mm, SD= 1.01) as 11% bigger max dilation size(6.76mm) than simple design signs. The size distribution for complex design signs (5.07), as shown in Table 11, increased by 6.7% over simple design signals (4.75).

Table 11. Description of pupillometry analysison tangent segment for two signs

	Size(SD)	ICA(SD)	25%	50%	75%	max
complex	4.52(1.01)	.06(.08)	3.97	4.50	5.07	6.76
simple	4.07(1.14)	.48(.61)	3.62	4.23	4.75	6.08

According to Figure 18, the increase in size remained constant over time. The ICA, however, did not show a strong behavior on tangent segments as shown in Figure 19.

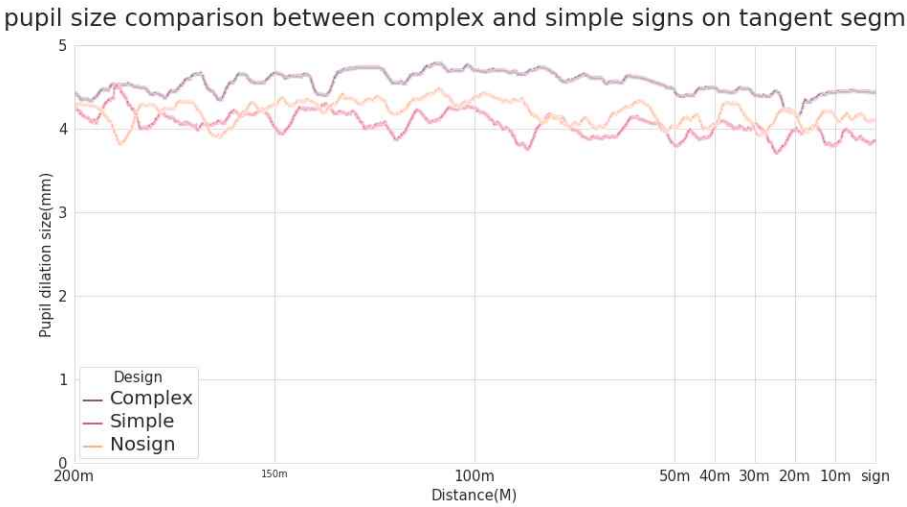


Figure 18. Pupil size comparison for two signs on tangent segment

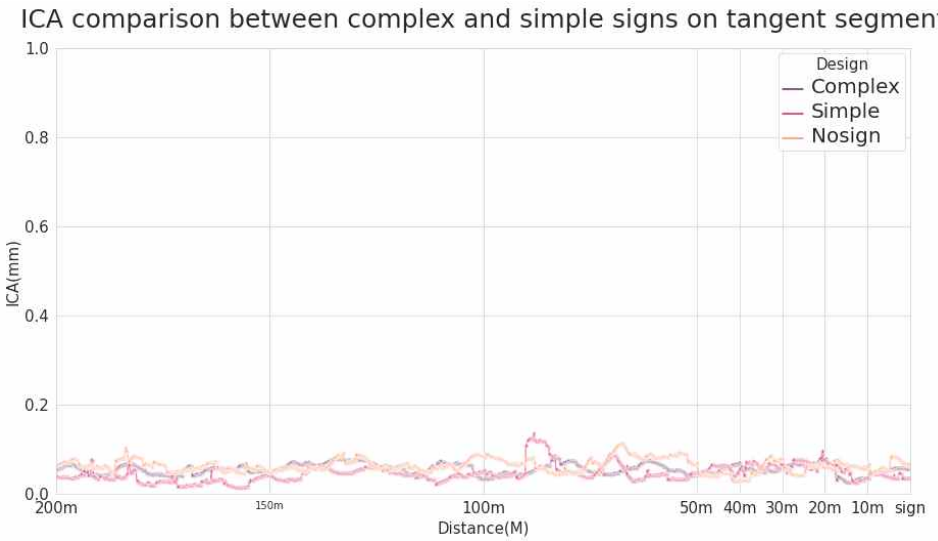


Figure 19. ICA comparison for two signs on tangent segment

3.2.1.2 Pupil dilation analysis for two signs on curve segment

In terms of pupilometry comparison of two signs on curve segment, complex design signs showed 3% increase in size(Mean=4.19, SD= 1.11). For an ICA, in particular, both complex and simple design signs showed a strong behavior on curve segment. complex sign(M=.19, SD=.71) showed 35% increased in ICA compare to simple design(M=.14, SD=.61) as Table 12 shown. For the size distribution, complex design(4.75) 5.7% increase in size than simple design(4.49) for 75% of data.

Table 12. Description of pupillometry analysis on curve segment for two signs

	Size(SD)	ICA(SD)	25%	50%	75%	max
complex	4.19(1.13)	.19(.71)	3.93	4.25	4.75	6.41
simple	4.03(1.05)	.14(.61)	3.84	4.16	4.49	7.41

According to Figure 20, pupil size comparisons for geometrical analysis are not significant. Meanwhile, Figure 21 suggests promising results of ICA analysis for two signs on curve segments.

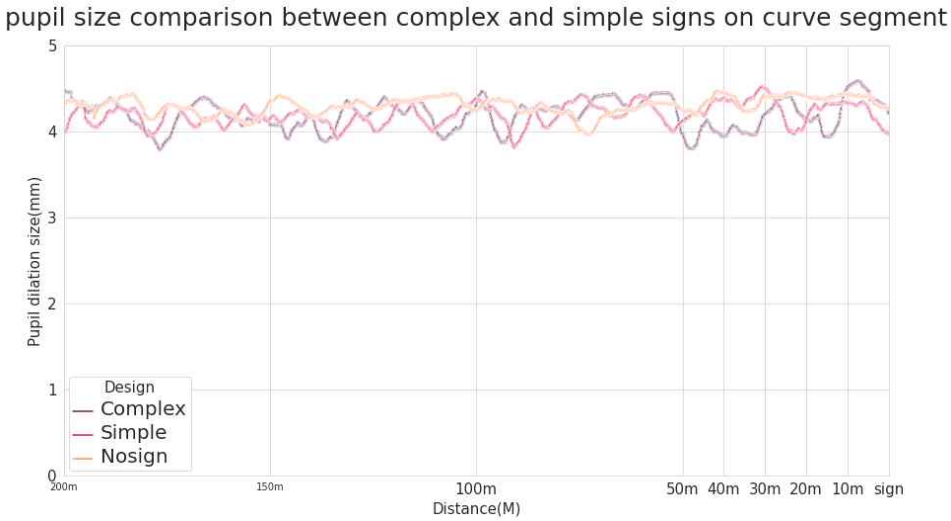


Figure 20. Pupil size comparison for two signs on curve segment

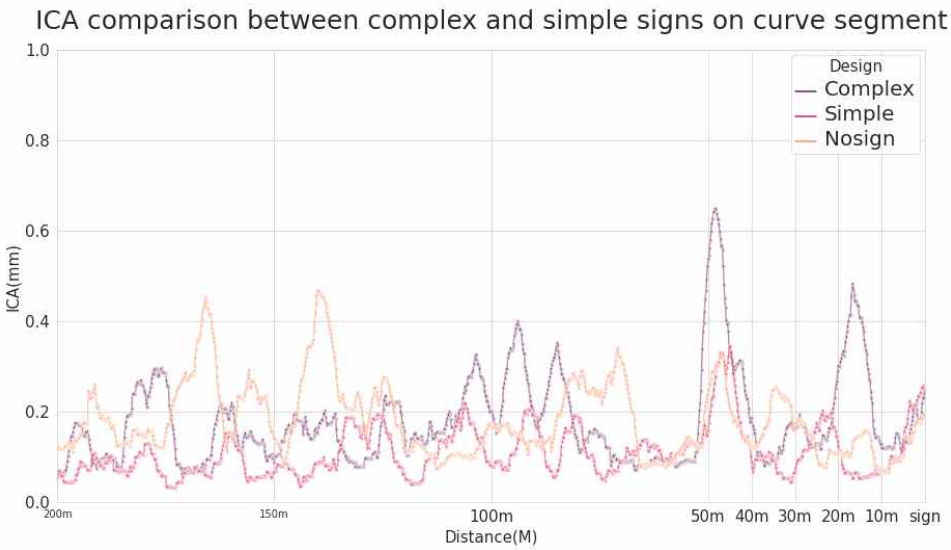


Figure 21. ICA comparison for two signs on curve segment

3.2.2 Straight Road vs. Curve Road

Based on the visual workload caused by the geometric structure, the pupil size of the tangent segment(M=4.14, SD=1.19) was smaller than the pupil size of the curve segment(M=4.30, SD=1.19) as Table 13 shown. In addition, the ICA of the curve segment(M=.17, SD=.56) recorded a higher value than that of the tangent segment(M=.07, SD=.09). Table 13 confirmed that the maximum pupil size(6.31) and 75% pupil size(4.74) were higher in the tangent segment than in the curve segment(max=6.21,75%=4.61), but 25% in the curve segment.

Table 13. Description of pupillometry analysis on two types of segment

	Size(SD)	ICA(SD)	25%	50%	75%	max
tangent	4.14(1.19)	.07(.09)	3.59	4.30	4.74	6.31
curve	4.30(.74)	.17(.56)	3.98	4.29	4.61	6.21

According to Figure 22, in the tangent segment, it is possible to observe many sections where pupil sizes are smaller than those of the curve segment. Similarly, according to Figure 23, the curve segment has a turbulent ICA graph compared to the tangent segment.

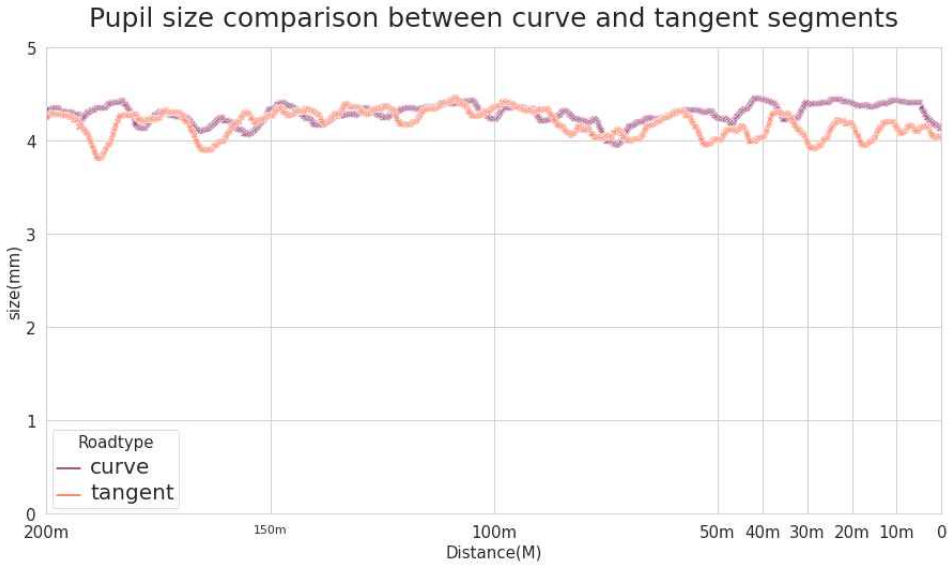


Figure 22. Pupil size comparison for tangent and curve segments

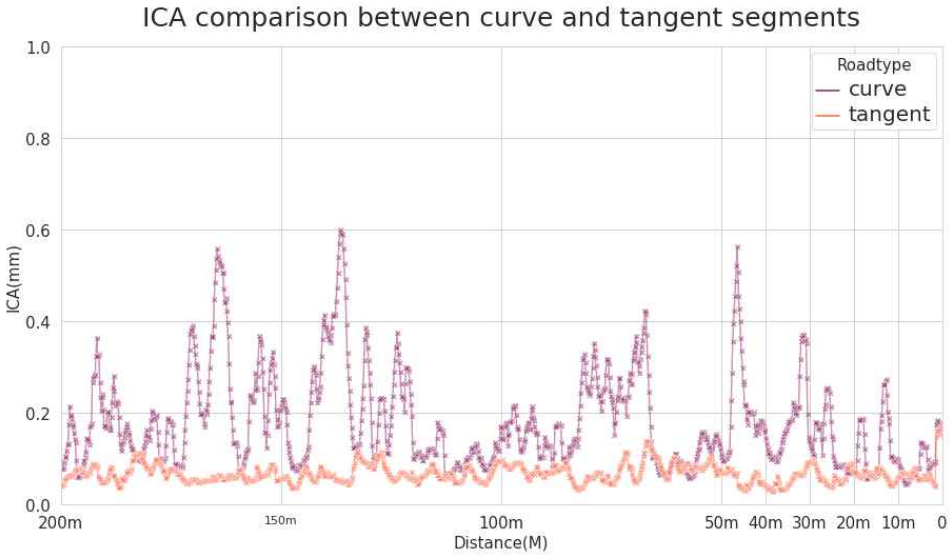


Figure 23. ICA comparison for tangent and curve segments

3.2.3 Underpass

A sign experiment confirmed the change in pupil size induced by visual demand, but the road terrain experiment did not. In order to see if the driver's psychological state can be confirmed through data by showing other changes in the driving environment, roads with a similar geometric structure were divided into above and underpasses and simulated while driving through the use of an underpass driving simulator.

It was determined that the pupil of the driver expanded upon entering the underpass to a valid value, and contracted when the lane change was made when it left the underpass. As Figure 24 and 25 shown, almost all subjects were influenced by underpass, and the majority of subjects agreed that underpass made their pupil dilation increased. The analytical results indicated that underpass had a significant effect on pupil size, ICA, and maximum size of pupil dilation as Table 14 shown. Furthermore, the ICA, which was checked while driving, confirmed that there was a valid difference in driving on an underground roadway as Figure 26 shown.

Table 14. Description of pupillometry analysis on abovepass and underpass

	Size(SD)	ICA(SD)	25%	50%	75%	max
above	3.97(1.47)	.31(1.03)	4.00	4.40	4.73	7.57
underpass	4.79(1.72)	.43(1.29)	4.65	5.34	5.71	8.74

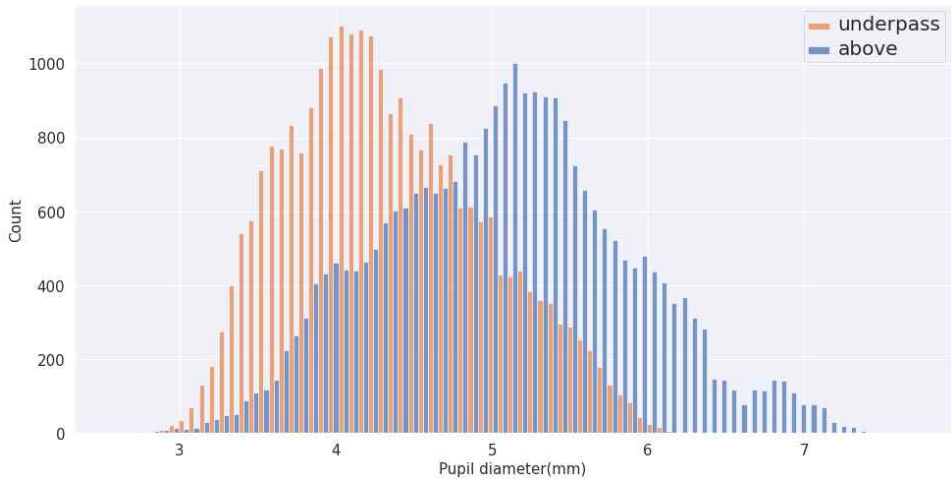


Figure 24. Histogram of pupil size on above and underpass

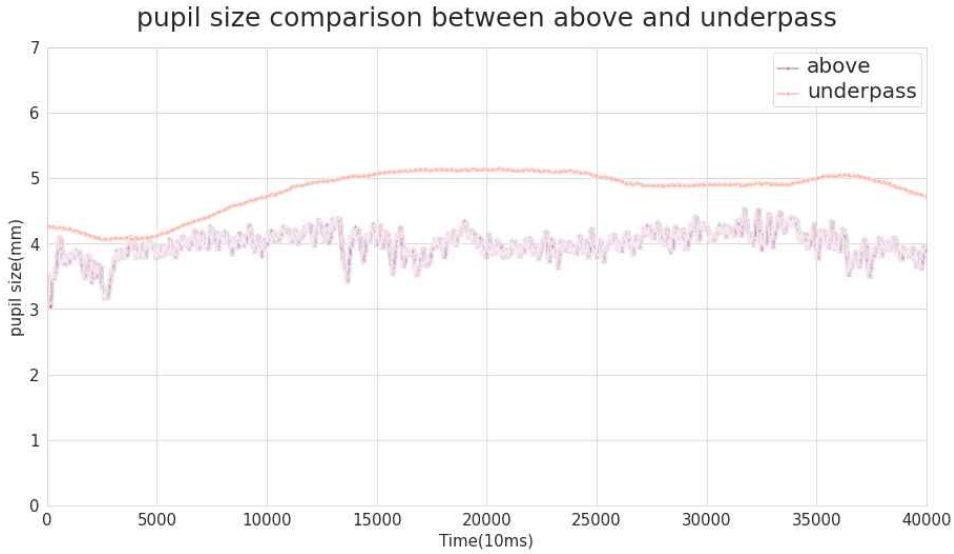


Figure 25. Pupil size comparison between above and underpass

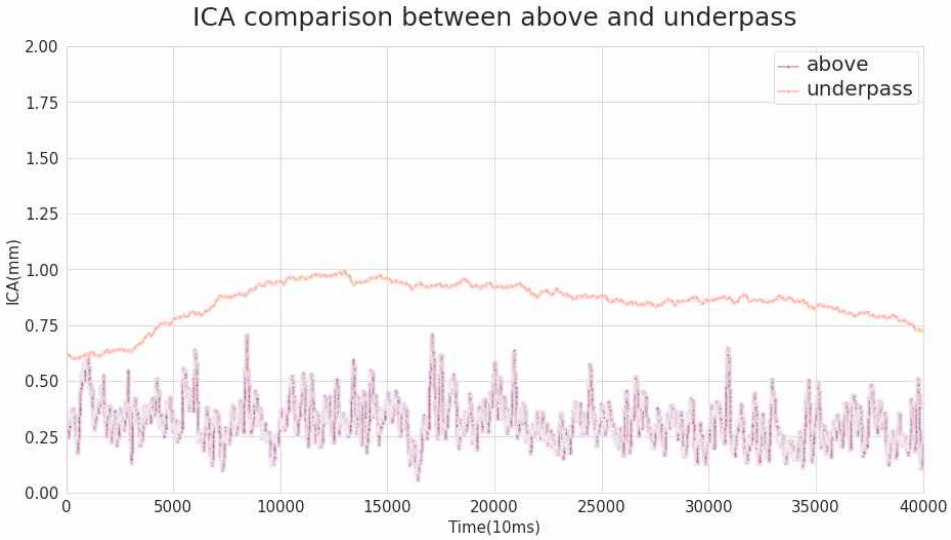


Figure 26. ICA comparison between above and underpass

When performing visual tasks on complex design signs, the primary visual performances for participants changed dramatically compared to complex design signs. Majority of participants' pupil dilation increased as well as ICA level. It was found that performing the visual tasks while driving on complex design sign significantly inc0reased the pupil size, increased pupil dilation deviation. A steady increase in pupillometry across all levels was observed.

Chapter 5 Conclusions

1. Summary

In this paper, ocular behavior assessment methods were evaluated and mental workload was investigated while driving in different roadway environments. Research using a driving simulator has been considered a valid tool for evaluating countermeasures related to driving behavior, since it was able to minimize confounding effects by controlling the various factors affecting driving behavior. A driving simulator study is typically safer and less expensive than a field study, and data collection is relatively easy. The results of the driving simulation study were analyzed objectively. Researchers recruited 37 subjects for the experiment; one was unable to participate due to nausea.

By analyzing previous studies, finding problems and selecting alternative indicators, a representative indicator is derived, and the best alternative is derived by using the target indicators, according to the driving environment. This has led to an optimal alternative based on the vision-workload analysis scale.

In terms of visual demand, experiment on two different design signs showed promising result of gaze behavioral analysis of driving vision-workload. the evaluation reflected driving vision-workload demanded by complex design signs. However, pupil datas showed inconsistent results on two signs. On tangent segment, pupil size didn't show any significant different yet ICA was found to be different. Due to the above, the signs were visible

from a long distance in a tangent section, so it attracted attention, and through the gaze data, the driver's vision workload could be determined. Meanwhile, on curve segment, pupil size was found to be different when ICA was not. Conclusively, one can suggest that visual demand impact driving vision-workload in different form regarding geographical property.

For geographical experiment, fixation rate and duration showed difference in number yet seemed weak compare to previous analysis results. However, tangent segment showed more vertically dispersed eye movement while curve segment showed more horizontally dispersed eye movement. The result suggest that curve demand more horizontal eye movement for the purpose of successful driving performance. Therefore, spatial gaze concentration was able to reflect driving vision-workload on geographical property.

In regard to driving environment, experiment on above and underpass showed promising result on pupillometry analysis of driving vision-workload. The evaluation reflected driving vision-workload demanded by cognitive workload by underpass. However, gaze behavioral analysis did not reflect driver's cognitive workload through fixation rate, duration and spatial gaze concentration. The result suggest that driving vision-workload related to an environment can only be detected through pupil dilation size and ICA.

Based on fixation rate, duration, and spatial gaze concentration,

gaze behavioral data effectively measure the visual workload caused by geography and signs while driving. Yet, geographical evaluation must be reviewed. However, driving on the underpass did not significantly change the gaze behavioral data.

The pupil dilation data did not measure driving workload as a result of the gaze-related visual demand generated by geometry, whereas the cognitive workload generated while driving on an underpass was largely reflected.

This finding supports that driving vision-workload is a multidimensional phenomenon, and illustrates the necessity of continuing research in a variety of driving environments in the future.

Table 15. Description of vision related measures evaluation details

	Visual Demand	Geometry	Environmental
Fixation visit	○	×	×
Fixation duration	○	×	×
Gaze Density	○	×	×
Gaze Horizontal Dispersion	○	○	×
Gaze Vertical Dispersion	○	○	×
Pupil size	×	×	○
ICA	×	○	○

2. Conclusions

This thesis evaluated the alternative, representative indicator for driving vision-workload with respect to vehicle operation. By analyzing the data collected in a driving simulator, the following conclusions can be reached:

- Under this experimental evironment, complex design signs increased all subject's number of fixation visit and duration time and range of spatial gaze concentration at the location of 200m from the signs.
- The analytical results indicated that spatial gaze concentration must regard both fixation density and distribution of duration time to acknowledge the true dispersion of attention while driving.
- According to literatures, the analytical results including blink rate and duration must be considered as an effective contribution.
- ICAs were effective in geometrical and environmental experiments.

the thesis searched for various visualization-related studies, and it provided a chance to explore different approaches to studying cognitive load and mental workload using visual data, as well as a number of ways to measure a particular workload during driving. By looking at the relevant research content, it seems that the study should be approached. A future study should analyze the appropriate analysis scale for the vision-workload based on various

driving environments, and determine which external factors affect the specific scale more than the other variables. Considering the aim of this study is to present and introduce analysis scales required to study visual overload during driving, it seems appropriate as a direction for future research to determine what the optimal analysis scale for different driving environments is appropriate.

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국문초록

기존 교통연구는 주행중 업무부하와 안전성에 관한 시각행태관련 연구를 다루지 않았다. 이에 본 논문은 주행중 시각행태관련 척도를 종속변수로 통합하려는 최근의 시도를 조사하였다. 이후 복잡한 적용방법을 고려하여 시각적 척도개발을 목적으로 한다.

문헌 검토는 상황인지능력을 필요로 하는 업무를 하는 동안 동공이 확장되고 눈을 깜빡이는 현상이 나타나며 확장과 눈을 깜빡이는 정도는 업무 난이도에 따라 달라짐을 논의한 연구 및 카메라와 적외선 조명기를 이용한 다양한 영상 기반 시각행태 측정방법이 논의되었다.

또한, 시각 연구가 시선 행동, 동공 측정 및 눈 깜빡임을 통해 인지 부하를 연구하는 방법을 소개했다. 각 연구에 대한 조사는 종합적인 문헌검토이 완료된 후 연구범위를 기준으로 수행되었다. 인지부하의 독립변수와 시각행동측정의 종속변수를 포함기준을 충족하도록 정량화를 시작으로 주행중 시각적 작업부하 분석척도를 선정했다. 제한된 기준을 고려하여 주행중 시각적 작업부하 측정을 목적으로하는 척도후보 목록을 도출했다.

이 연구는 정량적 척도 설계와 공간복합적 척도 설계를 구별하기 위해 주시점 방문횟수 및 주시시간을 고려해 주시분포를 확인했다. 다음으로, 본 논문에서는 기하구조적 작업부하 평가에서 동공행태 측정값과 주시점 행태변화 유효여부를 판단하기 위해 직선구간과 곡선구간 동공측정값을 비교하였다.

주행공간 환경에 대한 시각적 업무량에 대한 동공 측정 측정의 타당성을 결정하기 위해, 우리는 지하도와 지상일반 도로를 비교했다.

운전 시뮬레이터를 이용한 연구는 운전행동에 영향을 미치는 다양한 요인을 제어함으로써 방해요소를 최소화할 수 있었기 때문에 운전행동과 관련된 대책을 평가하는 데 유효한 도구로 평가되어 왔다. 주행 시뮬레이터

연구는 일반적으로 현장 연구보다 안전하고 비용이 적게 들고 데이터 수집이 비교적 용이하다. 주행 시뮬레이터 연구의 결과는 객관적으로 분석되었다. 실험은 37명의 실험자를 대상으로 진행되었고 한 명은 메스꺼움으로 제외되었다.

본 연구는 선행연구 검토 및 주행 시뮬레이터 실험을 통해 다차원 현상이론에 기초한 다양한 맥락에서 주행중 시각적 업무량을 분석, 정의, 측정 및 적용하기 위한 모듈러 프레임워크를 제시함으로써 학술적 의의를 갖는다.

주제어: 주행 시뮬레이터, 시각측정기, 주행중 업무량, 시각행태, 주행행태

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